

Inhibitory basal ganglia nuclei differentially innervate pedunculopontine nucleus subpopulations and evoke differential motor and valence behaviors

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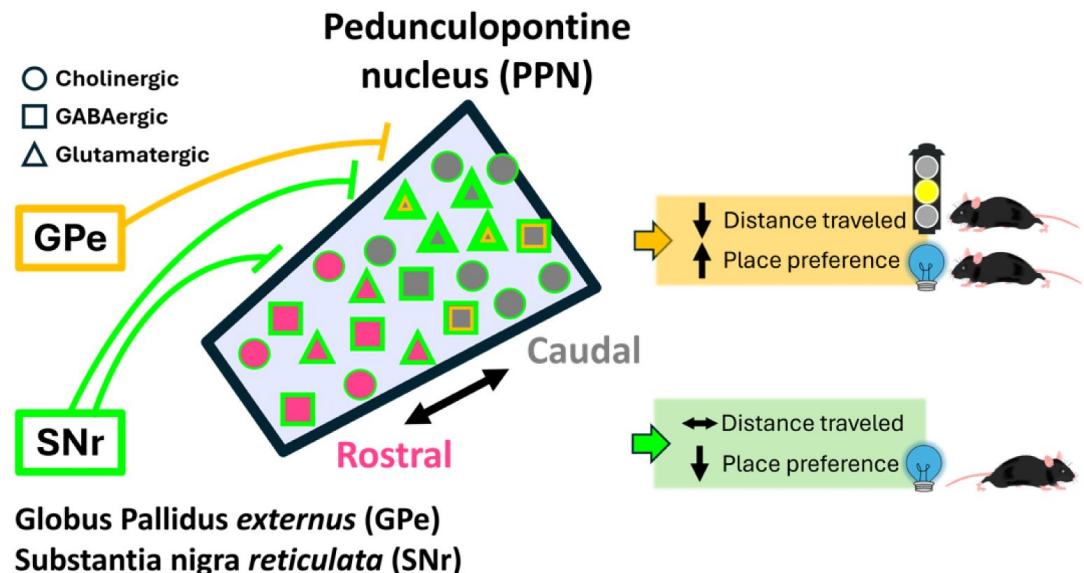
Fallah et al carefully dissect projections from substantia nigra pars reticulata (SNr) and the globus pallidus externa (GPe) - two key basal ganglia nuclei - to the pedunculopontine nucleus (PPN), a brainstem nucleus that has a central role in motor control. They consider inputs from these two areas onto three types of downstream PPN neurons - GABAergic, glutamatergic, and cholinergic neurons - and carefully map connectivity along the rostrocaudal axis of the PPN. Overall, this **important** study provides **convincing** data on PPN connectivity with two key input structures that will provide a basis for further understanding PPN function.

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Abstract

The canonical basal ganglia model predicts that the substantia nigra *pars reticulata* (SNr) and the globus pallidus *externa* (GPe) will have specific effects on locomotion: the SNr inhibiting locomotion and the GPe enhancing it. In this study, we use *in vivo* optogenetics to show that the GPe exerts non-canonical effects on locomotion while the SNr has no gross motor impact through inhibition of the PPN. We also show that these structures mediate opposing effects on reward. To understand how these structures differentially modulate the PPN, we use *ex vivo* whole-cell recording with optogenetics to comprehensively dissect the SNr and GPe synaptic connections to regionally- and molecularly-defined populations of PPN neurons. The SNr inhibits all PPN subtypes, but most strongly inhibits caudal glutamatergic neurons. The GPe selectively inhibits caudal glutamatergic and GABAergic neurons, avoiding both cholinergic and rostral cells. This circuit characterization reveals non-canonical basal ganglia pathways for locomotion and valence.

Graphical Abstract



Introduction

The pedunculopontine nucleus (PPN) is a brainstem structure heavily interconnected with subcortical structures, such as the basal ganglia, thalamus, and spinal cord. The PPN has been implicated in locomotor control and valence processing^{1,2,3,4}. However, understanding PPN circuitry is complicated by its anatomical and molecular heterogeneity. It has three molecularly-defined cell types, distinguished by their major neurotransmitter⁵ and distinct rostral and caudal topography. While cholinergic and glutamatergic neurons are more densely packed in the caudal PPN, GABAergic neurons are more densely packed in the rostral PPN^{5,6}. These molecularly-defined populations display very minimal overlap^{5,7,8}. This unique topography has been conserved across species^{9–11}, but it is unknown how the basal ganglia connects to each of the distinct cell types in the rostral and caudal PPN regions to modulate locomotor and valence outputs.

While the PPN interacts with all basal ganglia nuclei^{3,9}, the inhibitory inputs from the substantia nigra *pars reticulata* (SNr) and the globus pallidus *externus* (GPe) to the PPN are of particular interest because of their roles in the canonical motor output pathways for ‘stop’ and ‘go,’ respectively. Both the SNr and GPe are known to send axonal projections to the PPN^{12–18}. Previous studies have identified SNr and GPe monosynaptic connections to the PPN cholinergic and glutamatergic neurons using rabies tracing^{13–17}. However, inputs to the GABAergic neurons have only been described in terms of the entire mesencephalic locomotor region (MLR), which extends beyond the PPN¹⁴. While SNr projections to MLR GABAergic neurons were identified but cannot be isolated to the PPN, GPe projections to the MLR GABAergic neurons were not detected¹⁴. Of note, these previous studies consider the PPN as a whole rather than separating its rostral and caudal subregions. While viral tracing can provide insight into neuroanatomical connections, limitations of these techniques^{19–21} and evidence that the rostral and caudal PPN may have distinct functions^{22–25} prompt a need for electrophysiological studies to determine whether the SNr and GPe inputs differentially innervate regionally-defined PPN neurons.

Optogenetic manipulations of the molecularly-defined PPN cell types have resulted in variable, and sometimes contradictory, locomotor and valence behaviors^{7,14,16,17,22,33}. More recent studies have refined our understanding of these opposing behavioral effects by selectively stimulating the rostral or caudal GABAergic and glutamatergic PPN neurons^{22–25}. In the basal ganglia, direct stimulation of molecularly-defined SNr and GPe subpopulations can also have distinct effects on locomotion and valence^{34–42}. Despite the potential for the basal ganglia to modify behavior through the motor brainstem¹⁴, it is not clear how the SNr and GPe influence the rostral and caudal PPN to modify locomotion and valence.

Here, we use *ex vivo* whole-cell patch clamp electrophysiology paired with optogenetics to selectively stimulate the SNr or GPe axons while recording inhibitory inputs to the cholinergic, GABAergic, and glutamatergic PPN neurons to comprehensively characterize the strength and pattern of inhibition across the rostrocaudal PPN axis. We find that the GPe preferentially inhibits caudal GABAergic and glutamatergic PPN neurons whereas the SNr most strongly inhibits a caudal ‘hotspot’ of PPN glutamatergic neurons. Stimulating SNr or GPe axons over the PPN *in vivo* evokes opposing valence processing outcomes with place aversion during SNr stimulation and place preference during GPe stimulation. Surprisingly, and counter to the predictions of the canonical basal ganglia model, the GPe decreases locomotion through its projections to the PPN. While both the SNr and GPe inhibit the PPN, our results show that each nucleus differentially modulates the activity of specific cell types in the rostral and caudal PPN and are implicated in non-canonical basal ganglia circuits for modulating locomotion and valence processing.

Results

SNr and GPe axons display distinct distribution patterns across the rostral and caudal PPN

To characterize the axon distribution pattern from the SNr and GPe throughout the rostral and caudal extent of the PPN, we stereotactically injected a channelrhodopsin virus with a fluorescent marker (AAV1-hSyn-ChR2-eYFP) into either the SNr or GPe of ChAT-Cre/Ai9-tdTomato mice (**Figure 1A&C**). We sliced 200 μ m sagittal slices from mouse brains three weeks after viral injection, and cleared the brain slices using the CUBIC clearing method⁴³. Using the red fluorescent ChAT-positive neurons to define the borders of the PPN, we found that SNr axons appear in both the rostral and caudal PPN. The SNr axons fill the rostral PPN evenly, but appear more distinctly clustered in specific areas of the caudal PPN (**Figure 1B**), whereas GPe axonal projections more densely fill the caudal region and strikingly avoid the rostral region (**Figure 1D**). These anatomical findings predict that the SNr will inhibit both rostral and caudal PPN neurons, while the GPe will selectively inhibit the caudal PPN. However, this axon imaging method does not reveal whether these axons form functional synapses with the neurons in the PPN and does not provide information on the strength of inhibition from each structure onto specific PPN cell types. Therefore, to evaluate the functional connectivity and synaptic characteristics from these inhibitory basal ganglia output structures to regionally- and molecularly-defined PPN neural subpopulations, we used *ex vivo* whole-cell electrophysiology to systematically record the inhibitory input to the rostral and caudal PPN from the SNr and GPe in three genetic mouse lines (ChAT-cre, Vgat-cre, and Vglut2-cre) which label cholinergic, GABAergic, and glutamatergic PPN subpopulations, respectively (**Figure S1**).

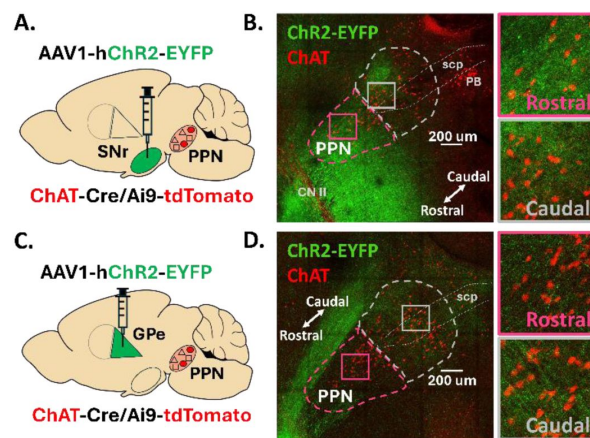


Figure 1.

SNr and GPe axons display distinct distribution patterns across the rostral and caudal PPN.

(A,C) Stereotaxic injection of AAV1 delivering *hSyn-ChR2-eYFP* to the SNr or GPe of ChAT-Cre/Ai9-tdTomato mice, respectively. (B, D) Confocal images of EYFP-filled SNr or GPe axons across the PPN, respectively. *CNII*: cranial nerve II, *scp*: superior cerebellar peduncle; *PB*: parabrachial nucleus

The SNr inhibits cholinergic neurons in both the rostral and caudal PPN to a similar extent, but evokes stronger rebound in rostral neurons

To evaluate the functional connection between the SNr and cholinergic PPN neurons, we performed *ex vivo* whole-cell patch clamp electrophysiology paired with optogenetics to record synaptic currents in ChAT-Cre/Ai9-tdTomato mice injected with Chr2 in the SNr three weeks prior (**Figure 2A**). Neurons were targeted for whole-cell patch clamp in both the rostral and caudal PPN across three sagittal slices to evaluate the medial to lateral extent of the PPN (**Figure 2B**). To measure optically-evoked inhibitory post-synaptic currents (oIPSCs), cholinergic PPN neurons were held at -50 mV in whole-cell voltage clamp and blue (470 nm) light was applied (2 ms pulse duration, 13 mW) to activate the Chr2-infected SNr axons. In connected neurons, inhibitory currents were recorded in the presence of glutamatergic receptor blockers (AP5, NBQX, CNQX) during stimulation (**Figure 2C**). In a subset of neurons, a GABA-a receptor blocker, GABAzine, was applied to confirm the oIPSCs were GABA-mediated (**Figure 2C**). GABAzine eliminated SNr-evoked inhibitory currents in most cells, but a small residual current remained in some cells (see *Methods* for details). We found that all rostral ($n=19/19$ cells, $N=6$ mice) and essentially all caudal ($n=26/27$, $N=6$) recorded cholinergic PPN neurons responded to SNr axon activation with an observable oIPSC (**Figure 2D**).

To determine whether the SNr input to the rostral and caudal PPN cholinergic neurons differed in synaptic strength, we measured the amplitude of each oIPSC in a 20 Hz train of 2 ms optical stimulations applied for 2 seconds. There was no difference in amplitude of the first oIPSC in the stimulus train between rostral vs caudal cholinergic neurons (mean $\pm$ SEM; first oIPSC $n=15$ Rostral: 92.6 ± 18 pA, $n=20$ Caudal: 115.2 ± 17 pA; unpaired t-test, $t=0.9041$, $df=33$, $p=0.3725$; **Figure 2E&F**). Going forward, the first oIPSC amplitude of the train will be referred to as the first oIPSC.

While the rostral and caudal anatomical separation of PPN neurons is commonly used, it is a relatively coarse division. Therefore, we carefully mapped the location of each recorded cholinergic PPN neuron by matching slices to their bregma reference in the Paxinos and Franklin's mouse brain atlas and aligning the midline of cholinergic neuron distribution which separates the loosely spread rostral neurons and densely packed caudal neurons (See *Methods* for details, **Figure 2G**). The strength of the SNr inhibitory connection is depicted by a color scale representative of the first oIPSC (**Figure 2G**, right). Throughout both the rostral and caudal PPN, cholinergic neurons receiving larger inputs from the SNr are intermixed with neurons receiving smaller inputs.

To compare short-term synaptic plasticity characteristics between rostral and caudal cholinergic PPN neurons, we normalized the amplitude of each current in the train to the first current. The oIPSC amplitude remained relatively constant with each subsequent pulse in both rostral and caudal cholinergic neurons (**Figure 2H**). In addition, the paired-pulse ratio (PPR) of the peak of the first two currents in the stimulation train did not differ (mean $\pm$ SEM; PPR $n=15$ Rostral: 0.96 ± 0.04 , $n=20$ Caudal: 0.97 ± 0.05 ; unpaired t-test, $t=0.1082$, $df=33$, $p=0.9145$; **Figure 2I**). SNr synaptic inputs onto cholinergic neurons do not undergo significant short-term synaptic plasticity in either the rostral or caudal PPN.

Although oIPSC amplitude is indicative of the strength of a synapse, current size does not always correlate with functional impact on a neuron's action potential output^{20,44,45}. Because cholinergic PPN neurons display spontaneous firing^{46–50}, we characterized the impact of SNr-mediated inhibition on action potential firing rate. While optically stimulating SNr axons with the same 20 Hz blue light protocol as above, we observed a decrease in firing rate in both rostral and caudal cholinergic PPN neurons (**Figure 2J–K**). However, there was no significant difference

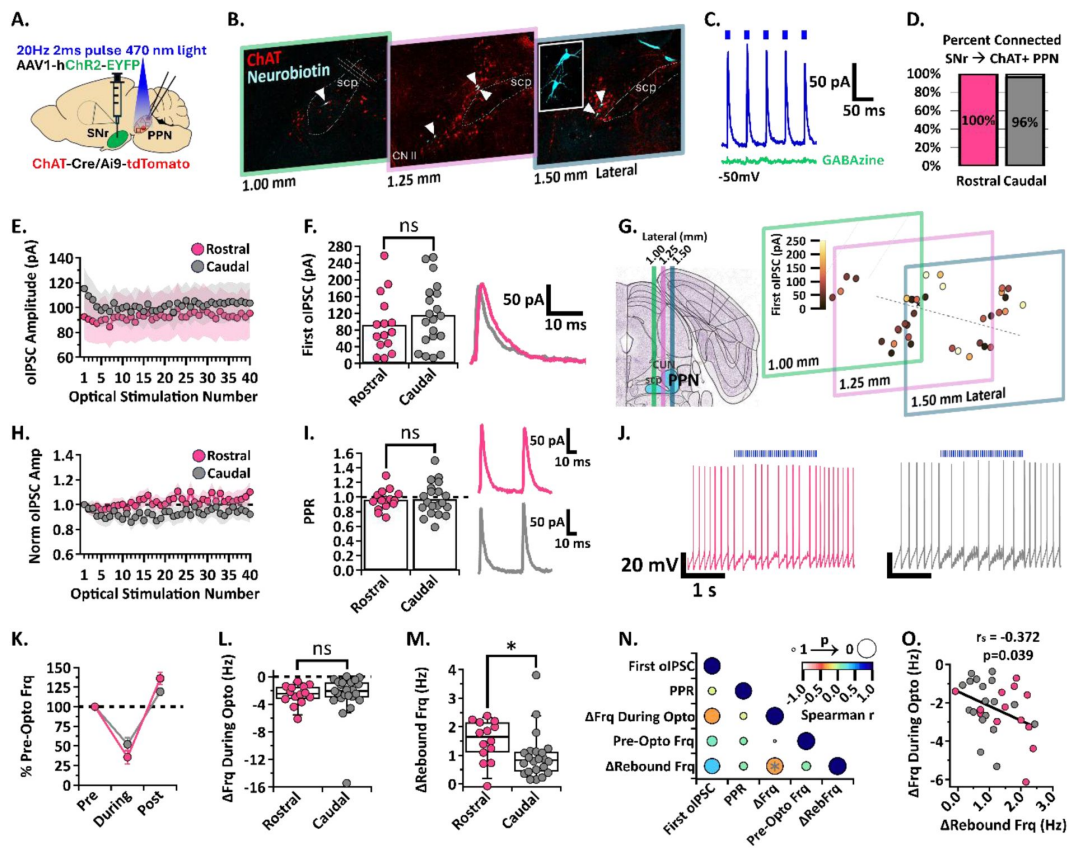


Figure 2.

SNr inhibition of rostral and caudal ChAT+ PPN neurons.

(A) Experimental set up to identify red ChAT+ PPN neurons for whole-cell patch clamp while stimulating ChR2-filled SNr axons [N=6]. (B) White arrowheads pointing to neurobiotin-filled patched neurons within the PPN across three 200 μ m slices. (C) Example trace of the first five oIPSCs [blue] in the 2-second 20 Hz train inhibited by GABA-a receptor blocker, GABAzine [green], while holding the cell at -50mV. (D) Percent connected among patched neurons in the rostral and caudal regions. (E) Average oIPSC amplitude at each of 40 optogenetic light pulses in n=15 rostral neurons and n=20 caudal neurons. (F) *Left*, Individual cell data for the first oIPSC amplitude and, *right*, example current traces. (G) Cell mapping of patched neuron locations with the first oIPSC amplitude represented by the color scale. (H) Normalized current amplitudes in E. (I) *Left*, Individual cell data for the PPR between the first two oIPSC amplitudes in the train and, *right*, example current traces. (J) Example voltage traces of action potential firing during a 2-second 20 Hz train stimulation in rostral (left) and caudal (right) neurons. (K) Percent of pre-optical stimulation firing frequency during stimulation and rebound in n=14 rostral vs n=23 caudal neurons. (L) Individual cell data for the absolute change in frequency during optical stimulation [Δ Frq During Opto]. (M) Individual cell data for the absolute change in rebound frequency post-stimulation [Δ RebFrq]; rostral vs. caudal p=0.0142. (N) Correlation analysis, color scale representing Spearman r [-1,1] and size representing p-value [1,0]. (O) Negative correlation between the absolute change in frequency during stimulation and post-stimulation rebound; $r_s = -0.372$, $p = 0.039$. * $p < 0.05$; bar graph data represent mean $\pm$ SEM; box plots show median line with boxes showing IQR and whiskers showing 9th and 91st percentiles.

between the percent frequency change in rostral and caudal neurons during SNr axon stimulation (mean $\pm$ SEM; %Frequency $n=14$ Rostral: During Stimulation 35.62 ± 8.87 %, Post Stimulation 136.1 ± 7.803 %; $n=23$ Caudal: During Stimulation 52.11 ± 8.43 %, Post Stimulation 119.1 ± 4.614 %; 2way ANOVA, Stimulation $_{F(1, 35)}=81.9$ $p<0.0001$, Region $_{F(1, 35)}=0.002052$ $p=0.9641$, Interaction $_{F(1, 35)}=3.273$ $p=0.0790$; **Figure 2K**) or in the absolute decrease in firing frequency during optical stimulation [median (IQR); Δ Frq During Opto $n=14$ Rostral: -2.47 Hz (-3.37 to -1.44 Hz), $n=23$ Caudal: -2.10 Hz (-3.11 to -0.84 Hz); Mann Whitney, $U=131$, $p=0.3602$; **Figure 2L**]. These data show that the SNr equally inhibits action potential firing in rostral and caudal PPN cholinergic neurons.

Post-inhibitory rebound firing has been observed in the PPN as well as other brainstem structures^{49,51,57} and is associated with greater excitability, temporal encoding, and generation of oscillatory activity⁵⁸. Indeed, we found that the rostral and caudal cholinergic PPN neurons exhibited an increase in action potential firing after inhibition (**Figure 2K**). This is in agreement with previous literature showing rebound spikes following inhibitory postsynaptic potentials recorded in PPN neurons of male rats during electrical stimulation of the SNr^{51,52}. However, rostral cholinergic neurons had greater rebound activity compared with caudal cholinergic neurons as measured by the absolute change in rebound frequency [median (IQR); Δ RebFrq $n=14$ Rostral: 1.64 Hz (1.00 to 2.17 Hz), $n=22$ Caudal: 0.83 Hz (0.42 to 1.14 Hz); Mann Whitney, $U=79$, $p=0.0142$; **Figure 2M**]. This aligns with previous findings showing that rostral cholinergic neurons tend to have stronger intrinsic rebound activity compared to caudal neurons⁴⁹. Importantly, we show that such rebound can be evoked by synaptic inhibition from the SNr.

To detect trends among SNr connectivity characteristics and intrinsic electrophysiological properties of cholinergic PPN neurons, we performed a correlation analysis to determine the strength and directionality of the relationship among five parameters [first oIPSC, PPR, Δ Frq During Opto, pre-optical stimulation frequency (Pre-Opto Frq), and Δ RebFrq; **Figure 2N**]. We found a weak but significant negative correlation between the frequency decrease during SNr axon stimulation and the rebound frequency increase (Spearman $r=-0.372$, $p=0.039$, **Figure 2N&O**). This finding is expected since strong inhibition is more effective at producing rebound activity than weak inhibition. However, rostral PPN cholinergic neurons do not receive significantly stronger inhibition from the SNr compared with caudal cholinergic neurons (**Figure 2F&L**). Therefore, larger rebound frequency in rostral neurons is likely due to differences in intrinsic cellular properties rather than synaptic properties⁴⁹. Put together, these findings show that the SNr inhibits rostral and caudal cholinergic PPN neurons to a similar extent, but evokes stronger rebound firing in the rostral neurons.

The SNr inhibits GABAergic neurons in both the rostral and caudal PPN to a similar extent

While there are several rabies tracing studies suggesting SNr axons project to the cholinergic and glutamatergic PPN neurons^{13,17}, SNr projections to the GABAergic PPN neurons are inferred from a study of the entire mesencephalic locomotor region¹⁴ and a characterization of their synaptic strength has not been conducted. Therefore, we evaluated the synaptic characteristics between the SNr and GABAergic neurons in the rostral and caudal PPN. In Vgat-Cre/Ai9-tdTomato mice, we injected Chr2 into the SNr to optically stimulate axons over the PPN while patching Vgat+ GABAergic PPN neurons (**Figure 3A**). We post-hoc stained the slices for ChAT+ neurons to verify the location of all patched GABAergic neurons within the cholinergic PPN. We found that all rostral ($n=12/12$, $N=6$) and most caudal ($n=20/22$, $N=6$) recorded GABAergic PPN neurons received inhibitory input from the SNr (**Figure 3B**).

Although there was no major difference in the proportion of GABAergic neurons receiving inhibitory input from the SNr in the rostral vs caudal PPN, we compared the current amplitudes to characterize the synaptic strength and pattern of this connection. The oIPSC amplitudes across the

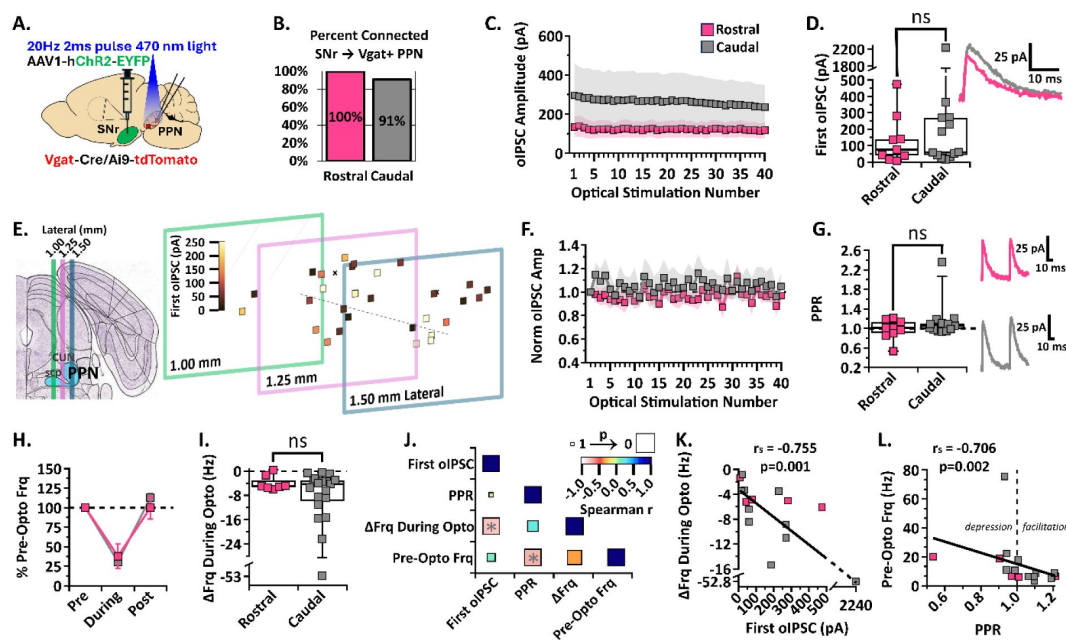


Figure 3.

SNr inhibition of rostral and caudal Vgat+ PPN neurons.

(A) Experimental set up to identify red Vgat+ PPN neurons for whole-cell patch clamp while stimulating ChR2-filled SNr axons [N=6]. (B) Percent connected among patched neurons in the rostral and caudal regions. (C) Average oIPSC amplitude at each of 40 optogenetic light pulses in n=9 rostral neurons and n=13 caudal neurons. (D) *Left*, Individual cell data for the first oIPSC amplitude and, *right*, example current traces. (E) Cell mapping of patched neuron locations with the first oIPSC amplitude represented by the color scale. (F) Normalized current amplitudes in C. (G) *Left*, Individual cell data for the PPR between the first two oIPSC amplitudes in the train and, *right*, example current traces. (H) Percent of pre-optical stimulation firing frequency [% Pre-Opto Frq] during stimulation and rebound in n=7 rostral and n=19 caudal neurons. (I) Individual cell data for the absolute change in frequency during optical stimulation [ΔFrq During Opto]. (J) Correlation analysis, color scale representing Spearman r [-1,1] and size representing p-value [1,0]. (K) Negative correlation between the absolute change in frequency during stimulation and first oIPSC amplitude; $r_s = -0.755$, $p = 0.001$. (L) Negative correlation between the pre-optical stimulation firing frequency and the PPR; $r_s = -0.706$, $p = 0.002$. Box plots show median line with boxes showing IQR and whiskers showing 9th and 91st percentiles.

2-second stimulation as well as the first oIPSC showed no difference between rostral and caudal GABAergic neurons [median (IQR); first oIPSC n=9 Rostral: 76.5 pA (27.0 to 209.9 pA), n=13 Caudal: 59.7 pA (37.7 to 269.7 pA); Mann Whitney, $U=50$, $p=0.6005$; **Figure 3C&D**]. We also did not observe significant differences in short-term synaptic plasticity between rostral and caudal GABAergic neurons [median (IQR); PPR n=9 Rostral: 1.01 (0.88 to 1.16), n=13 Caudal: 1.06 (0.96 to 1.15); Mann Whitney, $U=48$, $p=0.5123$; **Figure 3E F&G**]. Defining short-term depression as a $PPR < 0.95$ and short-term potentiation as $PPR > 1.05$, we found 3/9 rostral and 7/13 caudal GABAergic PPN neurons showed short-term potentiation, while 3/9 rostral and 3/13 caudal GABAergic PPN neurons showed short-term depression. These data suggest additional heterogeneity within the GABAergic PPN subpopulation. Similar to the SNr connection with cholinergic PPN neurons, SNr input to the GABAergic PPN does not result in significant short-term synaptic plasticity and does not differ in amplitude or characteristics between rostral and caudal regions.

Because GABAergic PPN neurons display spontaneous firing⁵⁰, we recorded action potential firing in current clamp while optically stimulating the SNr axons. The percent inhibition was not significantly different between rostral and caudal neurons (mean $\pm$ SEM; %Frequency n=7 Rostral: During Stimulation 37.91 ± 15.91 %, Post Stimulation 100.0 ± 14.57 %; n=19 Caudal: During Stimulation 30.68 ± 8.03 %, Post Stimulation 113.2 ± 5.187 %; 2way ANOVA, Stimulation $_{F(1, 24)}=44.63$ $p < 0.0001$, Region $_{F(1, 24)}=0.09642$ $p=0.7588$, Interaction $_{F(1, 24)}=0.8841$ $p=0.3565$; **Figure 3H**). There was also no difference in the absolute frequency change during optical stimulation between rostral and caudal neurons [median (IQR); Δ Freq During Opto n=7 Rostral: -5.03 Hz (-5.55 to -1.34 Hz), n=19 Caudal: -4.29 Hz (-11.02 to -2.94 Hz); Mann Whitney, $U=55$, $p=0.5336$; **Figure 3I**]. These data show that the SNr equally inhibits action potential firing in rostral and caudal GABAergic PPN neurons and does not evoke significant rebound firing in either neural population.

Overall, we found that the SNr inhibits the rostral and caudal GABAergic PPN neurons to a similar extent, but observed greater variability in synaptic strength and impact on neuronal action potential firing among caudal GABAergic neurons. One caudal subgroup responded with very small (<60 pA) amplitude currents and another subgroup responded with currents more than triple that amplitude, with one neuron responding with a 2.2 nA inhibitory current. We mapped the location of each recorded GABAergic neuron, as described above, and found no specific location corresponding with the neurons receiving stronger SNr inhibition across the PPN landscape (**Figure 3E**). This strong variability suggests that GABAergic PPN neurons may be divided into functionally distinct neural types that do not correspond to their rostrocaudal anatomical location. To determine if the neurons receiving larger inhibitory currents have distinct characteristics, we performed a correlation analysis (**Figure 3J**). As expected, we found that the absolute frequency reduction during inhibition is correlated with the first oIPSC amplitude recorded in a cell (Spearman $r=-0.755$, $p=0.001$; **Figure 3K**). Surprisingly, however, we found a significant negative correlation between PPR and the pre-optical stimulation frequency (Pre-Opto Frq, **Figure 3L**). Specifically, input to slower-firing neurons displayed greater short-term synaptic facilitation. This finding supports the idea that multiple functional neural subpopulations may be present within the GABAergic PPN population. Together, these results show that the SNr functionally inhibits GABAergic neurons across the rostrocaudal extent of the PPN, but suggests the presence of additional heterogeneity within the GABAergic PPN population.

The SNr differentially inhibits the rostral and caudal glutamatergic PPN neurons

Previous rabies tracing anatomical studies have identified SNr inputs to glutamatergic PPN neurons^{14,16,17}, but a full characterization of their synaptic strength and rostral and caudal connectivity has not been conducted. In Vglut2-Cre/Ai9-tdTomato mice, we injected Chr2 into the SNr to optically stimulate axons over the PPN while patching Vglut2+ glutamatergic neurons in the PPN (**Figure 4A**). While holding cells at -50mV and applying 470 nm blue light stimulation, neurons were identified as connected if oIPSCs were observed. Similar to our findings in

cholinergic and GABAergic neurons, we found that all rostral (n=19/19, N=6) and all caudal (n=28/28, N=6) recorded glutamatergic neurons received inhibitory input from the SNr (**Figure 4B** [↗](#)).

Comparing the oIPSC amplitudes across the 20 Hz train, we discovered that the caudal glutamatergic neurons receive larger inhibitory currents than rostral glutamatergic neurons (**Figure 4C** [↗](#)). The median first oIPSC measured in caudal glutamatergic neurons was also significantly larger than those recorded in rostral neurons [median (IQR); first oIPSC n=13 Rostral: 61.6 pA (19.2 to 113.9 pA), n=13 Caudal: 313.5 pA (143.8 to 729.1 pA); Mann Whitney, U=29, p=0.0035; **Figure 4D** [↗](#)]. After post-hoc staining for the cholinergic PPN and mapping the location of each recorded glutamatergic PPN neuron as described above, we identified a 'hotspot' of strong SNr inhibition in a medial-caudal group of glutamatergic neurons (**Figure 4E** [↗](#)). This finding shows that a subset of caudal PPN neurons preferentially receive exceptionally strong SNr input.

To determine whether the SNr input to the rostral and caudal PPN show differential short-term synaptic plasticity, we compared the normalized oIPSC amplitudes. We found that inhibitory inputs to the rostral glutamatergic neurons display short-term synaptic facilitation with amplitudes increasing in subsequent stimulations (**Figure 4F** [↗](#)). However, strong facilitation occurred in only a few neurons that drove the mean upward and there was no significant difference between the PPR observed in rostral and caudal glutamatergic neurons when activating SNr axons [median (IQR); PPR n=13 Rostral: 1.07 (0.86 to 1.38), n=13 Caudal: 0.92 (0.89 to 1.02); Mann Whitney, U=77, p=0.7241; **Figure 4G** [↗](#)].

Because glutamatergic PPN neurons spontaneously fire^{16,50}, we wanted to determine if larger inhibitory currents mediated larger decreases in the firing rate of caudal PPN neurons. Therefore, we recorded tonic action potential firing in glutamatergic PPN neurons while optically stimulating SNr axons (**Figure 4H** [↗](#)). Although inhibition measured as the percent of pre-optical stimulation firing frequency was not significantly different between rostral and caudal neurons (mean ± SEM; %Frequency n=13 Rostral: During Stimulation 32.62 ± 10.21 %, Post Stimulation 100.3 ± 6.113 %; n=17 Caudal: During Stimulation 24.71 ± 7.30 %, Post Stimulation 97.48 ± 2.228 %; 2way ANOVA, Stimulation_{F(1, 28)}=133.6 p<0.0001, Region_{F(1, 28)}=0.5214 p=0.4762, Interaction_{F(1, 28)}=0.1765 p=0.6776; **Figure 4I** [↗](#)), we found that the absolute frequency decrease was significantly larger in caudal glutamatergic neurons compared to rostral neurons during SNr stimulation [median (IQR); ΔFrq During Opto n=13 Rostral: -3.21 Hz (-12.21 to -1.95 Hz), n=17 Caudal: -14.08 Hz (-20.04 to -6.07 Hz); Mann Whitney, U=55, p=0.0197; **Figure 4J** [↗](#)]. To examine the discrepancy between the percent decrease and the absolute decrease in firing frequency, we compared the spontaneous firing frequency between rostral and caudal glutamatergic neurons in cells that had no holding current applied. We found that caudal glutamatergic neurons fire faster than rostral glutamatergic neurons [median (IQR); Spontaneous Frequency n=11 Rostral: 6.8 Hz (5.4 to 17.2 Hz), n=16 Caudal: 20.6 Hz (11.2 to 37.2 Hz); Mann Whitney, U=45, p=0.0343; **Figure 4K** [↗](#)]. Together, these findings show that fast-firing caudal glutamatergic PPN neurons receive larger SNr-mediated IPSCs and display greater decreases in their action potential firing.

Since we found that the caudal glutamatergic neurons receive larger inhibitory currents and display greater decreases in firing rate during SNr axon stimulation, we performed a correlation analysis to evaluate the strength of the relationship among the synaptic and firing characteristics (**Figure 4L** [↗](#)). Interestingly, there was a weak but significant positive correlation between the PPR and frequency change during stimulation showing that input displaying short-term synaptic depression occurs in neurons with greater decreases in firing frequency (Spearman r=0.486, p=0.030; **Figure 4M** [↗](#)). As expected, SNr-mediated oIPSCs with larger amplitudes are correlated with greater decreases in firing frequency during stimulation (Spearman r=-0.841, p<0.00001; **Figure 4N** [↗](#)). Greater pre-optical stimulation firing frequency is correlated with larger frequency decreases during stimulation (Spearman r=-0.791, p<0.0001; **Figure 4O** [↗](#)). This finding is expected because the fast-firing neurons can be more strongly inhibited in terms of absolute decrease in

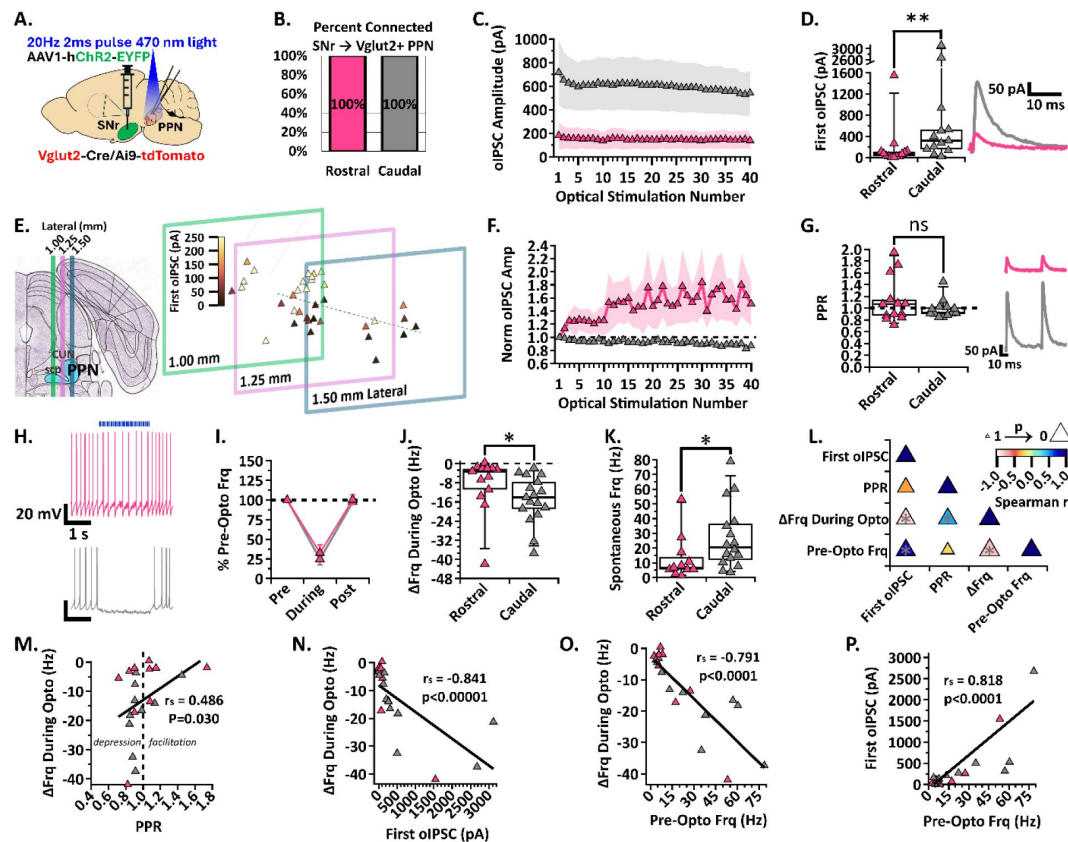


Figure 4.

SNr inhibition of rostral and caudal Vglut2+ PPN neurons.

(A) Experimental set up to identify red Vglut2+ PPN neurons for whole-cell patch clamp while stimulating ChR2-filled SNr axons [N=6]. (B) Percent connected among patched neurons in the rostral and caudal regions. (C) Average oIPSC amplitude at each of 40 optogenetic light pulses in n=13 rostral neurons and n=13 caudal neurons. (D) *Left*, Individual cell data for the first oIPSC amplitude and, *right*, example current traces; p=0.0035. (E) Cell mapping of patched locations with the first oIPSC amplitude represented by the color scale. (F) Normalized current amplitudes in C. (G) *Left*, Individual cell data for the PPR between the first two oIPSC amplitudes in the train and, *right*, example current traces. (H) Example voltage traces of action potential firing during a 2-second 20 Hz train stimulation in rostral and caudal neurons, top to bottom. (I) Percent of pre-optical stimulation firing frequency [% Pre-Opto Frq] during stimulation and rebound in n=13 rostral and n=17 caudal neurons. (J) Individual cell data for the absolute change in frequency during optical stimulation [Δ Frq During Opto]; p=0.0197. (K) Spontaneous frequency in n=11 rostral and n=16 caudal neurons; p=0.0343. (L) Correlation analysis, color scale representing Spearman r [-1,1] and size representing p-value [1,0]. (M) Positive correlation between the absolute change in frequency during stimulation and PPR, $r_s = 0.486$, $p = 0.030$. (N) Negative correlation between the absolute change in frequency during stimulation and first oIPSC amplitude; $r_s = -0.841$, $p < 0.00001$. (O) Negative correlation between the absolute change in frequency during stimulation and pre-optical stimulation frequency; $r_s = -0.791$, $p < 0.0001$. (P) Positive correlation between the first oIPSC amplitude and pre-optical stimulation frequency; $r_s = 0.818$, $p < 0.0001$. * $p < 0.05$, ** $p < 0.01$; box plots show median line with boxes showing IQR and whiskers showing 9th and 91st percentiles.

action potential frequency (i.e. a floor effect). Surprisingly, however, pre-optical stimulation firing frequency is correlated with the first oIPSC amplitude (Spearman $r=0.818$, $p<0.0001$; **Figure 4P**), indicating that the SNr is more strongly connected to glutamatergic PPN neurons with faster firing rates.

These correlations highlight strong selectivity in SNr projections to PPN glutamatergic neurons. Overall, these findings support that the SNr selectively targets and strongly inhibits fast-firing caudal glutamatergic PPN neurons compared to rostral neurons, and that there is a medial-caudal hotspot where SNr inhibition of PPN glutamatergic neurons is particularly strong.

The SNr most strongly inhibits caudal glutamatergic PPN neurons

Most of the previous PPN circuit work has been done with rabies tracing. Aside from technical limitations, rabies tracing can only indicate the presence of synaptic connections. Our electrophysiological findings comprehensively characterize the strength and characteristics of the SNr input to each of the PPN cell types in both the rostral and caudal regions. Using this comprehensive dataset, we compared SNr input across the different PPN cell types in each region under the same electrophysiological conditions. We found that the median amplitude of the first SNr-mediated oIPSC was the same among rostral PPN neurons irrespective of cell type [median (IQR); first oIPSC $n=15$ ChAT+: 74.7 pA (44.5 to 136.5 pA), $n=9$ Vgat+: 76.5 pA (27.0 to 209.9 pA), $n=13$ Vglut2+: 61.6 pA (19.2 to 113.9 pA); Kruskal-Wallis, $H_{(2,34)}=0.7002$ $p=0.7046$; **Figure 5A**].

The impact of these inhibitory inputs on neuronal firing did not differ with cell type as shown by the absolute change in frequency during SNr stimulation [median (IQR); Δ Frq During Opto $n=14$ ChAT+: -2.47 Hz (-3.37 to -1.44 Hz), $n=7$ Vgat+: -5.03 Hz (-5.54 to -1.34 Hz), $n=13$ Vglut2+: -3.21 Hz (-12.21 to -1.95 Hz); Kruskal-Wallis, $H_{(2,31)}=2.389$ $p=0.3028$; **Figure 5B**].

We found that the SNr-mediated oIPSCs in caudal glutamatergic neurons were significantly larger than in cholinergic and GABAergic caudal PPN neurons [median (IQR); first oIPSC $n=20$ ChAT+: 109.3 pA (58.5 to 168.6 pA), $n=12$ Vgat+: 59.7 pA (37.7 to 269.7 pA), $n=13$ Vglut2+: 313.5 pA (143.8 to 729.1 pA); Kruskal-Wallis, $H_{(2,43)}=8.717$ $p=0.0128$; Dunn's test, ChAT+ vs Vgat+ $p=0.6409$, ChAT+ vs Vglut2+ $p=0.0042$, Vgat+ vs Vglut2+ $p=0.0295$; **Figure 5C**].

Likewise, the impact of SNr inhibition on neuron firing rate was strongest in caudal glutamatergic neurons. The absolute frequency decrease during stimulation in caudal glutamatergic neurons was larger than both cholinergic and GABAergic neurons [median (IQR); Δ Frq During Opto $n=23$ ChAT+: -2.10 Hz (-3.11 to -0.84 Hz), $n=19$ Vgat+: -4.29 Hz (-11.02 to -2.94 Hz), $n=17$ Vglut2+: -14.08 Hz (-20.04 to -6.07 Hz); Kruskal-Wallis, $H_{(2,56)}=24.99$ $p<0.0001$; Dunn's test, ChAT+ vs Vgat+ $p=0.0049$, ChAT+ vs Vglut2+ $p<0.0001$, Vgat+ vs Vglut2+ $p=0.0332$; **Figure 5D**]. While we found no significant difference between SNr-mediated oIPSCs recorded in GABAergic and cholinergic neurons (**Figure 5C**), the GABAergic neuron firing frequency was more strongly inhibited than cholinergic (**Figure 5D**). The greater reduction in firing frequency may be driven by the caudal GABAergic subgroup receiving particularly large oIPSCs (**Figure 3D**) and could be explained if GABAergic PPN neurons have higher input resistance than cholinergic neurons, as has been previously suggested⁵⁹. Altogether, our findings show that the SNr most strongly inhibits the glutamatergic neurons followed by GABAergic neurons in the caudal PPN and more weakly inhibits rostral PPN neurons without any cell type bias (**Figure 5E**).

The GPe preferentially, but weakly, inhibits a subset of caudal GABAergic and glutamatergic PPN neurons

GPe projections to the cholinergic and glutamatergic PPN neurons have been shown in previous rabies tracing studies^{14,17}, but projections to the GABAergic MLR neurons was not detected¹⁴. To evaluate the synaptic strength of these connections and determine regional connectivity, we repeated the previous whole-cell patch clamp experiments paired with

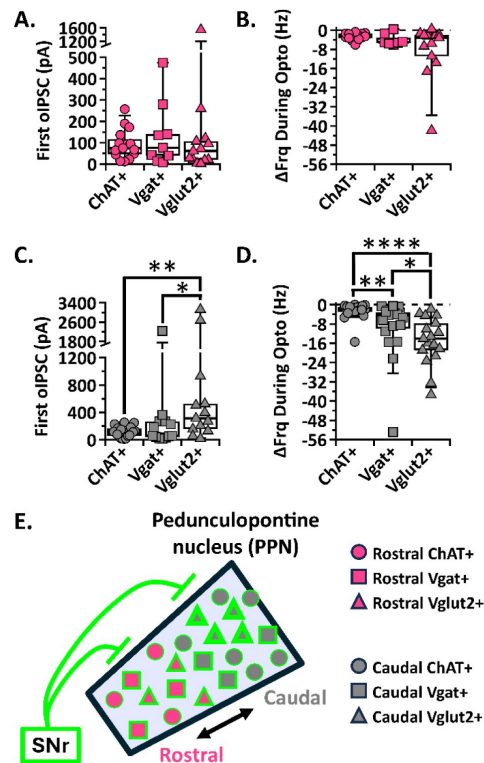


Figure 5.

The SNr most strongly inhibits caudal glutamatergic PPN neurons.

(A,C) Individual cell data for the first oIPSC amplitude recorded in each cell type for rostral and caudal PPN neurons, respectively. (B,D) Individual cell data for the absolute change in frequency during stimulation in each cell type for rostral and caudal PPN neurons, respectively. (E) Graphical depiction of SNr stimulation results. * $p < 0.05$, ** $p < 0.01$, **** $p < 0.0001$; box plots show median line with boxes showing IQR and whiskers showing 9th and 91st percentiles.

optogenetics, but this time injecting ChR2 into the GPe of ChAT-Cre/Ai9-tdTomato, Vgat-Cre/Ai9-tdTomato, and Vglut2-Cre/tdTomato mice (**Figure 6A**). If oIPSCs were observed in response to optical stimulation, the neuron was identified as connected. In a subset of neurons, a GABA-a receptor blocker, GABAzine, was applied to confirm the oIPSCs were GABA-mediated (See *Methods* for details; **Figure 6B**). We recorded GPe-mediated oIPSCs in all three cell types of the PPN. As predicted by the low density of GPe axons present in the rostral PPN (**Figure 1D**), very few rostral PPN neurons received synaptic input from the GPe. This was true within each molecularly-defined PPN cell type [caudal vs rostral % connected; ChAT+: 30% vs 0% (n=9/30 vs 0/7), Vgat+: 68% vs 38% (n=21/31 vs 6/16), Vglut2+: 69% vs 26% (n=24/35 vs 5/19); **Figure 6Cii–iii**]. These findings show that the GPe preferentially targets caudal PPN neurons compared to rostral neurons. Since so few rostral PPN neurons received GPe inhibition, we focused our characterization of GPe inhibition to the caudal PPN.

The percentage of connected caudal neurons was greater than the percentage of connected rostral neurons in both GABAergic and glutamatergic neurons (one-sided Fisher's exact test; Vgat: OR=0.2857, CI=0.09190 to 0.9391, p=0.047; Vglut2: OR=0.1637, CI=0.05069 to 0.5767, p=0.0033; **Figure 6Cii & iii**), but we found no statistical difference between the rostral and caudal connectivity in cholinergic neurons (OR=0, CI=0 to 1.301, p=0.1150; **Figure 6Ci**). The number of connected caudal cholinergic neurons was not significantly greater than its 0% rostral connectivity (**Figure 6Ci**). In addition, the average oIPSC amplitude across the train and the first oIPSC measured in the connected cholinergic neurons was small [median (IQR); first oIPSC n=6 ChAT+: 22.6 pA (9.75 to 50.20 pA); **Figure 6D&E**]. Among recorded caudal cholinergic PPN neurons, both connected and non-connected, these small inhibitory currents displayed negligible effects on the neuronal firing frequency of the cholinergic PPN population in both the percent of pre-optical stimulation frequency and absolute frequency change [median (IQR); n=29 ChAT+: %Frequency During Stimulation 96.67 % (85.1 to 100.7 %); Δ Frq During Opto -0.16 Hz (-0.90 to 0.03 Hz); **Figure 6H&I**]. These findings indicate that the GPe has essentially no direct effect on the cholinergic PPN neural population. For this reason, we exclude GPe inhibition of cholinergic PPN neurons in the synaptic characterizations shown in **Figures 6F & H**. Together, we show that the GPe preferentially inhibits caudal GABAergic and glutamatergic neurons while largely avoiding rostral and cholinergic PPN neurons.

To determine whether the synaptic strength and characteristics from the GPe to the PPN differed between GABAergic and glutamatergic PPN subtypes, we evaluated the amplitude of the GPe-mediated oIPSCs during the 2-second 20 Hz optical stimulation. Both GABAergic and glutamatergic neurons responded to GPe stimulation with oIPSCs ranging from ones to hundreds of picoamperes (**Figure 6D**). While there was no significant difference in the first oIPSC measured in GABAergic and glutamatergic neurons [median (IQR); first oIPSC n=19 Vgat+: 24.77 pA (6.72 to 54.17 pA), n=15 Vglut2+: 65.71 pA (14.50 to 133.4 pA); Kruskal-Wallis, $H_{(2,37)}=3.050$, p=0.2176; **Figure 6E**], the normalized amplitudes show that GPe input to GABAergic neurons facilitates with subsequent stimulations, while the input to glutamatergic neurons depresses (**Figure 6F**). The GPe-mediated inhibitory input to glutamatergic neurons displayed a significantly lower PPR than to GABAergic neurons [median (IQR); PPR n=19 Vgat+: 0.97 (0.76 to 1.2), n=15 Vglut2+: 0.81 (0.55 to 0.93); Mann-Whitney, U=76, p=0.0206; **Figure 6G**]. While short-term synaptic plasticity patterns were variable among GABAergic neurons with some depressing and others facilitating, GPe input to glutamatergic neurons generally depresses.

Short-term synaptic depression can dampen the impact that repeated IPSCs have on action potential output. Therefore, we compared the firing frequency change in each cell type during optogenetic stimulation of GPe axons. The PPN GABAergic neuron population firing rate was decreased to about 58% of their pre-stimulation firing with a median frequency decrease of 1.92 Hz during stimulation (**Figure 6H&I**). Similarly, the glutamatergic neuron population firing rate decreased to about 64% of their pre-stimulation firing with a median frequency decrease of 0.89 Hz during stimulation (**Figure 6H&I**). We found no statistical difference between GPe inhibition

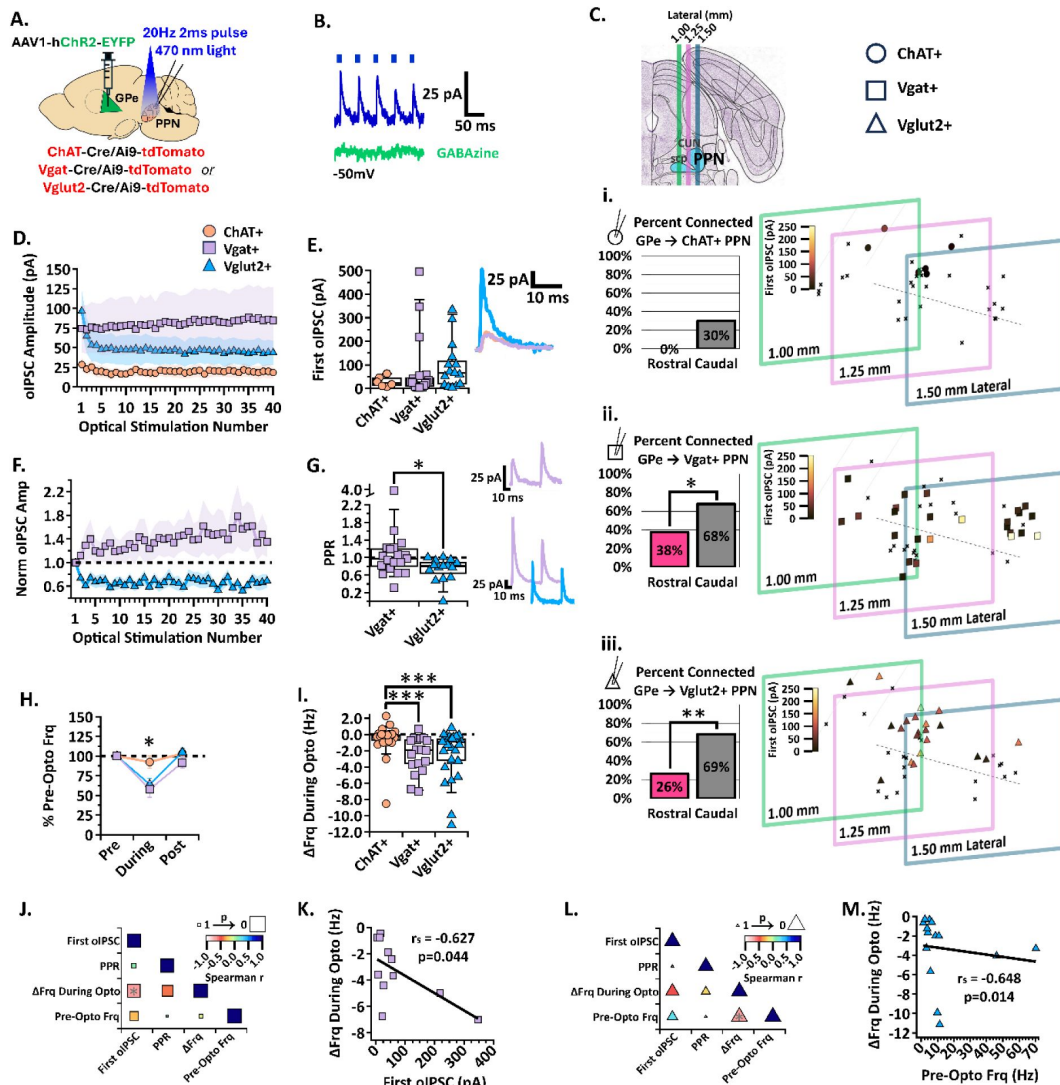


Figure 6.

GPe inhibition of the three PPN cell types.

(A) Experimental set up to identify red ChAT+, Vgat+, and Vglut2+ PPN neurons for whole-cell patch clamp while stimulating ChR2-filled GPe axons [N=6]. (B) Example trace of the first five oIPSCs in the 2-second 20 Hz train [blue] inhibited by GABA_A receptor blocker, GABAzine [green], while holding the cell at -50mV. (C) *Left*, Percent connected among patched neurons in the rostral and caudal regions and, *right*, cell mapping of patched locations with the first oIPSC amplitude represented by the color scale. *Top to bottom*, i. ChAT+, ii. Vgat+, and iii. Vglut2+ datasets. (D) Average oIPSC amplitude at each of 40 optogenetic light pulses in n=6 ChAT+, n=19 Vgat+, and n=15 Vglut2+ caudal PPN neurons. (E) *Left*, Individual cell data for the first oIPSC amplitude and, *right*, example current traces. (F) Normalized current amplitudes in C. (G) *Left*, Individual cell data for the PPR between the first two oIPSC amplitudes in the train; p=0.0206. *Right, top*, example current trace of short-term synaptic facilitation in Vgat+ neurons. *Right, bottom*, example current traces of short-term synaptic depression in Vgat+ and Vglut2+ neurons. (H) Percent of pre-optical stimulation firing frequency [%Pre-Opto Frq] during and post-stimulation in n=25 ChAT+, n=18 Vgat+, and n=29 Vglut2+ caudal PPN neurons. (I) Individual cell data for the absolute change in frequency during stimulation [Δ Frq During Opto].

(J) Correlation analysis for Vgat+ neurons, color scale representing Spearman r [-1,1] and size representing p-value [1,0]. (K) Negative correlation between the absolute change in frequency during stimulation and first oIPSC amplitude; $r_s = -0.627$, $p = 0.044$. (L) Correlation analysis for Vglut2+ neurons.

(M) Negative correlation between the absolute change in frequency during stimulation and the pre-stimulation firing frequency; $r_s = -0.648$, $p = 0.014$. *p<0.05, **p<0.01; box plots show median line with boxes showing IQR and whiskers showing 9th and 91st percentiles.

of firing frequency in caudal GABAergic and glutamatergic PPN neurons (mean $\pm$ SEM; %Frequency $n=29$ ChAT+: During Stimulation 92.39 ± 7.59 %, Post Stimulation 104.10 ± 5.23 %; $n=18$ Vgat+: During Stimulation 57.70 ± 10.11 %, Post Stimulation 91.81 ± 6.98 %; $n=24$ Vglut2+: During Stimulation 64.15 ± 7.63 %, Post Stimulation 104.7 ± 5.39 %; 2way ANOVA, Stimulation $_{F(1, 69)}=25.46$ $p<0.0001$, Cell type $_{F(1, 69)}=4.444$ $p=0.0153$, Interaction $_{F(1, 69)}=2.805$ $p=0.0674$; Fisher's LSD, $df=138$, %Frequency During Stimulation: ChAT+ vs. Vglut2+ $p=0.0035$, ChAT+ vs. Vgat+ $p=0.0016$, Vglut2+ vs. Vgat+ $p=0.5378$, %Frequency Post Stimulation: ChAT+ vs. Vglut2+ $p=0.8883$, ChAT+ vs. Vgat+ $p=0.2838$, Vglut2+ vs. Vgat+ $p=0.2184$; median (IQR); Δ Frq During Opto $n=25$ ChAT+: -0.16 Hz (-0.90 to 0.03 Hz), $n=18$ Vgat+: -1.92 Hz (-3.85 to -0.60 Hz), $n=29$ Vglut2+: -0.89 Hz (-3.28 to -0.43 Hz); Kruskal-Wallis, $H_{(2,69)}=17.21$, $p=0.0002$; Dunn's test, ChAT+ vs Vgat+ $p=0.0002$, ChAT+ vs Vglut2+ $p=0.0009$, Vgat vs Vglut2+ $p=0.4020$; **Figure 6H&I**). These findings show that the GPe inhibits GABAergic and glutamatergic neural populations to a similar extent.

To determine if there are significant relationships among the synaptic and firing characteristics, we performed a correlation analysis. Among GABAergic neurons receiving GPe inhibition, we found that the amplitude of the first oIPSC is significantly correlated with the absolute change in frequency during stimulation (Spearman $r=-0.627$, $p=0.044$, **Figure 6K**). Although GPe input to GABAergic neurons exhibited various PPRs, the correlation analyses revealed no significant relationship between PPR and absolute change in frequency during stimulation (**Figure 6J**). These findings suggest that the initial oIPSC amplitude of GPe input to GABAergic PPN neurons has a stronger relationship with firing rate than short-term synaptic plasticity does.

As nearly all GPe inputs to glutamatergic neurons exhibited short-term synaptic depression, we expected the correlation analysis to reveal a significant relationship between PPR and the change in firing frequency. However, similar to the GPe input to GABAergic neurons, PPR was not predictive of impact on neuronal activity (**Figure 6L**). GPe inhibition of the glutamatergic neurons showed a significant correlation between the absolute change in frequency during stimulation and pre-stimulation firing frequency (Spearman $r=-0.648$, $p=0.014$, **Figure 6M**). However, because the inhibitory synaptic current amplitude did not correlate with firing frequency, this correlation is likely due to a floor effect in which faster firing neurons have a greater capacity for reductions in firing frequency. Together, these findings show that the GPe preferentially inhibits caudal GABAergic and glutamatergic PPN neurons.

***In vivo* activation of GPe and SNr axons in the PPN show differential effects on locomotion and valence**

Once we established that the SNr and GPe inhibit different profiles of regionally- and molecularly-defined subpopulations of the PPN using *ex vivo* optogenetic electrophysiology, we wanted to evaluate the behavioral consequences of selectively stimulating these inputs *in vivo*. Direct optogenetic activation of PPN subpopulations have been previously shown to either promote or inhibit motion, sometimes with contradictory results.^{14,16,17,22,28} To determine whether differential inhibitory inputs onto the PPN influenced locomotor activity, we bilaterally injected wild-type mice with Chr2 in either the SNr or GPe and implanted an optical fiber over the PPN (**Figure 7A&B**).

Control mice were injected with an eGFP virus and implanted. After three weeks, we placed each mouse in the open field to measure gross locomotor behavior with and without optical stimulation of the SNr or GPe axons. We tracked the distance traveled for each mouse at baseline and during bilateral stimulation of either the SNr or GPe by applying blue light (473 nm, 4-4.5 mW) at 20 Hz with 2 ms pulses (4% duty cycle), for 1 minute at a time (**Figure 7C**). We found that stimulation of these two basal ganglia inputs to the PPN resulted in differential motor behaviors. Specifically, stimulation of SNr axons in the PPN increased distance traveled, while stimulation of GPe axons in the PPN decreased distance traveled [mean $\pm$ SEM; $N=9$ (3M, 6F) Control (Ctrl): 2.22 ± 0.24 m, $N=8$

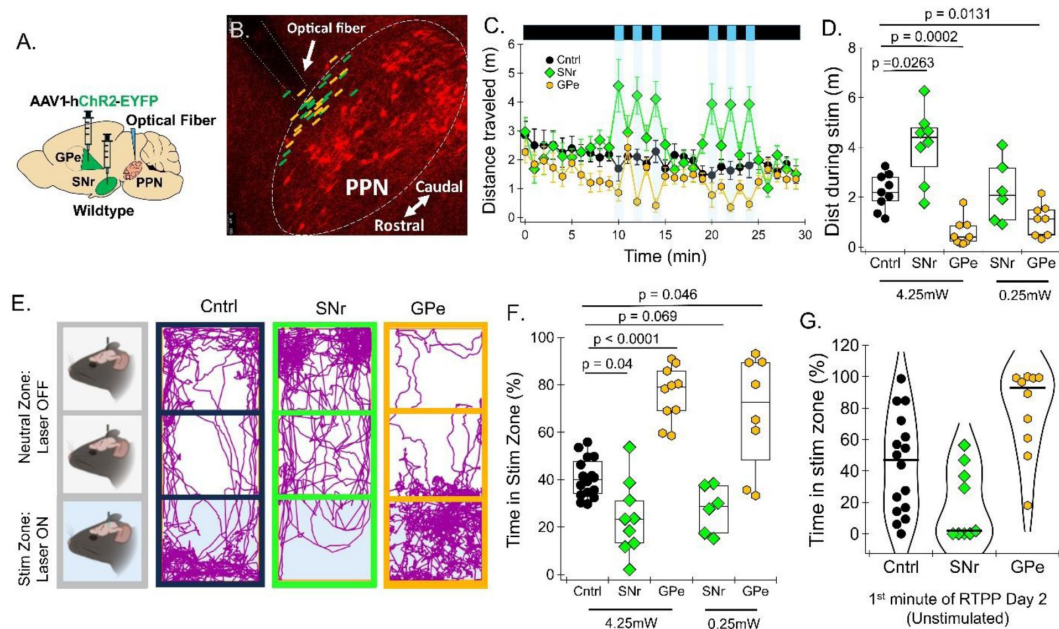


Figure 7.

***In vivo* activation of GPe and SNr axons in the PPN show differential effects on locomotion and opposite effects on valence.**

(A) Experimental set up to stimulate ChR2-filled SNr or GPe axons over the PPN *in vivo*. (B) Representative image of optical fiber tract overlaid with the approximate optical fiber placement for SNr-[green] and GPe-[orange] injected mice. (C) Distance traveled over time in an open field with 1 minute 20 Hz 4.25 mW optical stimulations over the PPN in N=9 control (Ctrl) mice (black circles), N=8 mice injected with ChR2 in the SNr (green diamonds), and N=9 mice injected with ChR2 in the GPe (orange hexagons); vertical blue lines represent periods of optical stimulation. (D) Average distance traveled for each mouse across the six 1-minute optical stimulations in the high (4.25 mW) and low laser power (0.25 mW) conditions. (E) Representative mouse track tracings during real time place preference task in a three-chamber box and continuously stimulating EGFP- or ChR2-filled axons over the PPN at 20 Hz in SNr- and GPe-injected mice when the mice are in the stimulation zone. (F) Percent time spent in the stimulation zone in N=16 control mice, N=9 mice injected with ChR2 in the SNr, and N=10 mice injected with ChR2 in the GPe with 4.25 or 0.25 mW laser power. (G) Percent time spent in the stimulation zone during the first minute reintroduced to the RTPP box on day 2 of RTPP with the laser off (no chamber is stimulated). Black line = median. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, **** $p < 0.0001$; box plots show median line with boxes showing IQR and whiskers showing 9th and 91st percentiles. See related supplemental Figures S2-S4 and Videos S1-S4.

(5M, 3F) SNr: 4.11 ± 0.51 m, N=9 (4M, 5F) GPe: 0.57 ± 0.18 m; Welch ANOVA $p < 0.0001$; Dunnett's test, Ctrl_{4.25mW} vs SNr_{4.25mW} $p = 0.0263$, Ctrl_{4.25mW} vs GPe_{4.25mW} $p = 0.0002$; **Figure 7D** & **Figure S2**, see Video S1&2].

The PPN has been implicated in reward processing and stimulation of the PPN can be reinforcing^{7,28,29,31,35,60}. Therefore, we evaluated the effect of GPe or SNr axon stimulation in the PPN on valence in the real-time place preference (RTPP) task. In a three-chamber apparatus, the mouse could freely move between chambers for ten minutes. Optical stimulation of the SNr or GPe axons in the PPN was applied when the mouse entered the stimulated chamber and remained on at 20 Hz (4% duty cycle) until the mouse exited that chamber (**Figure 7E**). To reduce the motor effects of optical stimulation, this experiment was conducted with unilateral stimulation only. Interestingly, we again found opposite effects when stimulating the two basal ganglia inputs to the PPN. Mice avoided the stimulated chamber when the SNr axons were stimulated, but preferred the stimulated chamber when the GPe axons were stimulated (mean $\pm$ SEM; N=16 (6M, 10F) Ctrl: 40.76 ± 2.0 %, N=9 (5M, 4F) SNr: 23.99 ± 5.2 %, N=10 (5M, 5F) GPe: 76.11 ± 3.7 %; Welch ANOVA, $p < 0.0001$; Dunnett's test, Ctrl_{4.25mW} vs SNr_{4.25mW} $p = 0.0437$, Ctrl_{4.25mW} vs GPe_{4.25mW} $p < 0.0001$, Ctrl_{4.25mW} vs SNr_{0.25mW} $p = 0.0695$, Ctrl_{4.25mW} vs GPe_{0.25mW} $p = 0.0467$; **Figure 7F**, see Video S3&4). To further control for potential motion confounds, we recorded the location preference of the mice in the absence of optical stimulation the day after the stimulated RTPP assay (regardless of whether they received low or high power on day 1). The mice were placed in the central, neutral chamber and their location was tracked for 10 minutes. During the first minute of this assay, 8 out of 9 mice who had SNr optical stimulation on the first day continued avoiding the stimulated chamber while 9 out of 10 mice which had GPe optical stimulation continued to prefer the previously stimulated chamber compared to the median percent time spent in the stimulated chamber for mice in the control condition [median (IQR); N=16 (6M, 10F) Ctrl: 47% (14.79 to 69.24 %), N=9 (5M, 4F) SNr: 1.833% (0 to 41.75 %), N=10 (5M, 5F) GPe: 92.83% (58 to 99.21 %); Kruskal-Wallis, $p = 0.0008$; Dunn's test, Ctrl vs SNr $p = 0.1433$, Ctrl vs GPe 0.0312; **Figure 7G**).

In addition to sending projections to the PPN, the GPe and SNr send axon collaterals to multiple other brain areas^{12,37,61,62}. Therefore, it is possible that stimulating these axons in the PPN could cause antidromic stimulation of the GPe and SNr cell bodies, resulting in the inhibition of non-PPN brain regions. However, it has recently been shown that 0.25 mW laser power stimulation of axons can prevent antidromic activation of cell bodies⁴². Therefore, we repeated these optogenetic behavioral experiments using 0.25 mW laser power. We found that the locomotion effect of SNr stimulation over the PPN was no longer present during low power stimulation, but GPe stimulation still significantly decreased locomotion (mean $\pm$ SEM; distance traveled N=9 (3M, 6F) Ctrl: 2.22 ± 0.24 m, N=6 (3M, 3F) SNr: 2.24 ± 0.50 m, N=8 (4M, 4F) GPe: 1.09 ± 0.22 m; Welch ANOVA $p < 0.0001$; Dunnett's test, Ctrl_{4.25mW} vs SNr_{0.25mW} $p > 0.999$, Ctrl_{4.25mW} vs GPe_{0.25mW} $p = 0.0131$; **Figure 7D**). In the RTPP task, SNr axon stimulation continued to evoke place aversion at the trend level and GPe axon stimulation continued to significantly evoke place preference (mean $\pm$ SEM; % time in stimulation zone N=16 (6M, 10F) Ctrl: 40.76 ± 2.1 %, N=6 (3M, 3F) SNr: 27.72 ± 4.0 %, N=8 (4M, 4F) GPe: 68.41 ± 8.5 %; Welch ANOVA, $p < 0.0001$; Dunnett's test, Ctrl_{4.25mW} vs SNr_{0.25mW} $p = 0.0695$, Ctrl_{4.25mW} vs GPe_{0.25mW} $p = 0.0467$; **Figure 7F**). These findings show that decreased locomotion and place preference are mediated by GPe inhibition of the PPN while aversion is mediated by SNr inhibition of the PPN.

The increase in locomotion observed when stimulating the SNr axons in the PPN with high laser power is unexpected given that previous studies directly stimulating SNr neurons report decreases in movement^{34,36}. Because injecting constitutive AAVs into the SNr can infect both GABAergic neurons and adjacent dopaminergic neurons of the substantia nigra pars compacta (SNc), we wanted to determine whether stimulating dopaminergic axons in the PPN enhanced locomotion and evoked aversion. To test this, we injected a cre-dependent ChR2 into the substantia nigra of DAT-cre mice (3M heterozygotes and 3F homozygotes) and ran open field and RTPP behavioral

assays. We found that selective stimulation of dopaminergic axons in the PPN increased locomotion at high laser power (**Figure S3A**) and had no effect on locomotion at low power (**Figure S3C**). This finding suggests that dopaminergic neurons in the SNc send collateral projections to both the PPN and other locomotor-related structures (e.g. the striatum) such that stimulation of PPN-projecting SNc neurons increases locomotion.

The effect of dopaminergic axon stimulation did not differ from whole substantia nigra stimulation in either high (mean $\pm$ SEM; N=8 (5M, 3F) SNr_{4.25mW}: 4.107 ± 0.51 m, N=6 (3M, 3F) DAT-Cre_{4.25mW}: 3.778 ± 0.28 m; unpaired t-test, $t=0.5159$, $df=12$, $p=0.6153$; **Figure S3B**) or low laser power (mean $\pm$ SEM; N=6 (3M, 3F) SNr_{0.25mW}: 2.238 ± 0.58 m, N=9 (3M, 3F) DAT-Cre_{0.25mW}: 1.984 ± 0.30 m; unpaired t-test, $t=0.4350$, $df=10$, $p=0.6728$; **Figure S3C**). On the other hand, we found that stimulation of dopaminergic axons in the PPN did not cause place aversion, but rather induced real-time place preference at both high and low laser power (**Figure S3D-F**). Selectively stimulating dopaminergic axons induced place preference behavior significantly different from and opposite to whole substantia nigra stimulation at high (mean $\pm$ SEM; N=9 (5M, 4F) SNr_{4.25mW}: 23.99 ± 5.18 %, N=6 (3M, 3F) DAT-Cre_{4.25mW}: 70.83 ± 10.06 %; unpaired t-test, $t=4.545$, $df=13$, $p=0.0005$; **Figure S3E**) and low laser power (mean $\pm$ SEM; N=6 (3M, 3F) SNr_{0.25mW}: 27.72 ± 4.0 %, N=6 (3M, 3F) DAT-Cre_{0.25mW}: 50.82 ± 4.8 %; unpaired t-test, $t=3.697$, $df=10$, $p=0.0041$; **Figure S3F**). Together, these findings show that the nigral increase in locomotion is not mediated through the PPN, and that dopaminergic input from the SNc to the PPN is rewarding while GABAergic input from the SNr to the PPN is aversive.

Discussion

In this study, we comprehensively characterize the synaptic strength and impact on neuron action potential firing in molecularly- and regionally-defined PPN subpopulations while stimulating the SNr or GPe using *ex vivo* electrophysiology and optogenetics. We also stimulated the SNr or GPe axons in the PPN *in vivo* which revealed that the SNr and GPe evoke differential locomotion and valence processing behaviors in non-canonical basal ganglia circuits. Together, these findings show that distinct and selective inhibition of PPN subpopulations by the SNr and GPe can alter behavioral output.

Region- and cell type-specific inputs to the PPN

The SNr and GPe have been previously shown to form monosynaptic synapses with PPN neurons using rabies viral tracing.^{12,17} While SNr input to cholinergic and glutamatergic PPN neurons has been reproduced across different studies,^{13,17} SNr inputs to GABAergic PPN neurons has only been inferred from studies of the entire mesencephalic locomotor region (MLR), consisting of both the PPN and cuneiform nucleus.¹⁴ Several studies have used slice electrophysiology to show that the SNr inhibits PPN neurons generally³² or to compare SNr input to cholinergic and non-cholinergic PPN neurons.^{48,51,52} However, a full characterization of the SNr input to regionally- and molecularly-defined PPN neurons in adult animals has been lacking. Using *ex vivo* whole-cell patch recordings, we were able to measure inhibitory currents in each of the PPN cell types during SNr stimulation showing that the SNr functionally inhibits each cell type within the rostral PPN to a similar extent while most strongly inhibiting a medial-caudal ‘hotspot’ of glutamatergic neurons.

While the GPe has been shown to project to the cholinergic and glutamatergic PPN neurons,^{15,17} our electrophysiological results show that GPe inhibition of the cholinergic neurons is weak to non-existent and its inhibition of the glutamatergic PPN neurons is caudally biased. In contrast to a previous study showing that the GPe does not project to GABAergic MLR neurons,¹⁴ we find that the GPe also preferentially inhibits a caudal subgroup of GABAergic PPN neurons. While the GPe selectively inhibits a caudal subpopulation of GABAergic and glutamatergic PPN neurons, its

inhibition is much weaker than SNr inhibition of the PPN in terms of the proportion of connected PPN neurons, the inhibitory current amplitudes, and the inhibition of neuronal firing. Overall, our findings utilize a systematic approach to characterize the synaptic strength between these inhibitory basal ganglia nuclei and the PPN across its rostrocaudal axis for each cell type allowing us to compare SNr and GPe to PPN circuitry with regional-level granularity. These results also encourage future work involving the PPN to consider the individual influence of its rostral and caudal regions.

Noncanonical GPe motor circuits

We found that stimulation of GPe axons over the PPN decreased locomotion in both low (putatively local) and high (putatively generating antidromic activity) laser power stimulations. This behavioral outcome appears counter to the canonical model of basal ganglia movement pathways, in which GPe activation increases movement. Although there is strong support for this canonical model, recent *in vivo* recordings have found subpopulations of GPe neurons that show activity patterns counter to this model (neurons that increase activity during immobility)^{63,64}. Similarly, a subset of GPe neurons paradoxically increase activity upon indirect pathway striatal projection neuron stimulation⁴¹. While one explanation for these heterogeneous responses could be local inhibition within the GPe^{40,41,65,66}, another possibility is that distinct subpopulations within the GPe differentially modulate locomotor behavior^{39–41,61,67}. Our findings support non-canonical basal ganglia motor pathways involving GPe neurons that project to the PPN.

Our characterization of region-specific inhibitory inputs to the PPN helps us understand how GPe inhibition of the PPN can mediate decreases in locomotion. Our electrophysiology experiments show that the GPe exerts minimal influence on the rostral PPN neurons which appear to decrease locomotion^{22,24,25} and preferentially inhibits caudal GABAergic and glutamatergic PPN neurons. Because stimulating the caudal GABAergic and glutamatergic PPN neurons increases locomotion²³, GPe selective inhibition of these caudal neurons could explain our behavioral experiments showing that stimulation of GPe axons over the PPN decreases locomotion. Future studies are needed to determine if GPe inhibition of both cell types is necessary for mediating decreased locomotion. Aligned with our findings, selectively stimulating the Npas1+ or FoxP2+ subpopulations in the GPe decreases locomotion^{39,41}. Some groups have found that direct stimulation of the PV+ GPe neurons increases locomotion^{39,62}; however, one study shows that inhibition of PV+ GPe neurons can increase locomotion⁴². While we show that GPe projections to the PPN decrease locomotion, future studies are needed to determine which GPe subpopulations project to the PPN to modulate locomotion.

GABAergic vs dopaminergic effects of nigral projections to the PPN

While we found that stimulating nigral axons in the PPN with high laser power increased locomotion, we observed no locomotor effect when using low laser power stimulation. The simultaneous inhibition of caudal PPN neurons that promote movement²³ and rostral PPN neurons that suppress movement^{22,24,25} may explain a net zero effect on locomotion during low power optical stimulation of SNr axons in the PPN. Indeed, low laser power stimulation of nigral axons in the most rostral implant location resulted in increased locomotion, while the most caudal implant location resulted in decreased locomotion (**Figure S4**). Interestingly, the enhanced locomotion observed during high laser power stimulation suggests that a subset of nigral neurons projecting to the PPN promote movement. However, the SNr canonically suppresses movement and the direct somatic stimulation of PV+ and GAD2+ SNr neurons that send axon collaterals to the PPN^{12,68} decreases gross locomotor movement^{34,68}. Since injecting constitutive AAVs into the substantia nigra can infect both GABAergic neurons of the SNr and adjacent dopaminergic neurons of the SNc, it is possible that stimulation of dopaminergic activity may mask the impact of stimulating the GABAergic SNr alone.

Both the increased locomotion with high laser power and the lack of change with low laser power was recapitulated when we ran these experiments while selectively stimulating dopaminergic projections from the SNc to the PPN (**Figure S3**). Therefore, our findings suggest that activation of the PPN-projecting dopaminergic SNc neurons promotes locomotion, but not through direct actions at the PPN. This is consistent with previous literature showing that SNc dopaminergic neurons send axons to both the PPN and striatum⁶⁹ and that direct stimulation of nigrostriatal dopaminergic neurons can increase movement initiation and vigor⁷⁰. Therefore, antidromic stimulation of the SNc may evoke dopamine release into the striatum to increase movement in our high laser power experiments. Future studies are needed to evaluate the motor behavioral impact of the different SNr cell types when inhibiting the entire PPN versus the rostral or caudal PPN subregions.

By contrast, we find that both high and low power non-selective SNr axon stimulation in the PPN is aversive (**Figure 7E-G**), while both high and low power selective stimulation of dopaminergic SNc axons in the PPN is rewarding (**Figure S3**). These findings are particularly interesting because they indicate that the dopaminergic vs GABAergic inputs from the substantia nigra to the PPN serve opposite functions regarding valence signaling. Importantly, these findings indicate that GABAergic input from the SNr is highly aversive and that this aversive signaling occurs through the PPN. Interestingly, these findings also show that dopaminergic SNc neurons do not require antidromic stimulation to signal reward. This means that dopaminergic signaling within the PPN may serve to reinforce behavior. Future work is needed to parse out specific reward and aversion pathways involving neural subpopulations within the SNr, SNc, and PPN.

Differential PPN inhibition by the SNr and GPe mediate opposing valence processing outcomes

We found that stimulation of SNr axons over the PPN was aversive in real time place preference (RTPP), while stimulation of GPe axons over the PPN led to a preference for the stimulated chamber. Previous studies directly stimulating SNr and GPe subpopulations have implicated these structures in reward and aversion processing.

Inhibition of Vgat+ SNr neurons has been shown to increase place preference³⁶ while its direct stimulation has no effect^{34,36}. However, selective stimulation of the PV+ SNr neurons induces aversion in RTPP³⁴. Although the GPe has been predominantly studied in the context of movement, recent work has begun to reveal a role for the GPe in valence processing^{42,71–73}. One recent study shows that inhibition of PV+ GPe neurons induced aversion while inhibition of Npas1+ GPe neurons induced place preference in RTPP⁴². Because activation of PV+ SNr neurons is aversive³⁴ and inhibition of PV+ GPe neurons is aversive⁴², it is likely that our results are due to activation of the PV+ SNr and GPe axons in the PPN. While we show that activation of the PPN-projecting SNr and GPe subpopulations strongly affect valence processing, future studies are needed to determine whether these effects are specifically mediated by PV+ SNr and GPe projections to the PPN.

Neurons in the PPN have been implicated in reward and aversive-related behaviors^{7,28,29,31–33,35,60}. The cholinergic PPN neurons are implicated in positive valence with place preference and increased lever pressing through their projections to the ventral tegmental area (VTA)^{28,29}. Aligned with our findings, direct inhibition of the cholinergic PPN neurons evokes place aversion²⁸. Of particular significance, we show that the SNr inhibits the cholinergic PPN neurons while the GPe axons avoid cholinergic PPN neurons. Therefore, it is possible that the difference between the SNr and GPe effects on valence is due to the difference in their inhibition of PPN cholinergic neurons.

Direct stimulation of the glutamatergic PPN neurons has also been implicated in positive valence with increased reinforcement behavior⁷. Both the cholinergic and glutamatergic PPN neurons directly influence dopamine release in the striatum and dopaminergic neuron activity in the SNc and VTA^{7,28,29,74–78}. Specifically, the caudal PPN neurons are thought to innervate the medial part of the SNc and the VTA^{9,78}, two major reward pathway hubs. Our data show that the SNr most strongly inhibits the caudal glutamatergic PPN neurons. Therefore, strong SNr inhibition of these neurons would remove excitatory drive from the reward-related dopaminergic neurons. This pathway could also contribute to the aversive effect of SNr axon stimulation in the PPN. Future experiments are needed to test whether direct inhibition of the caudal glutamatergic PPN neurons receiving SNr input is sufficient to evoke aversion.

The pathways underlying the GPe-mediated rewarding effect are less clear and may indicate functional heterogeneity among caudal PPN neurons. We find that the GPe only inhibits a subset of caudal GABAergic and glutamatergic PPN neurons. It is possible that the subset of GABAergic PPN neurons targeted by the GPe disinhibits local cholinergic and glutamatergic neurons to increase excitation of the rewarding dopaminergic neuron pathways. Another possibility is that the specific GPe-inhibited PPN population may have particularly aversive-properties. For example, PPN neurons that excite amygdala nuclei evoke place avoidance^{35,60}. In this case, GPe inhibition of these neurons could be rewarding by removing excitation of amygdala nuclei involved in negative valence processing. To understand the PPN circuitry involved in GPe-mediated place preference, future experimental work must determine whether inhibiting GPe-targeted GABAergic or glutamatergic PPN subpopulations alone is sufficient to evoke place preference.

Conclusions

We have systematically characterized SNr and GPe inputs across the rostrocaudal axis of the PPN for each cell type – cholinergic, GABAergic, and glutamatergic. We show that the SNr inhibits nearly every PPN cell recorded with differential strength determined by cell type and anatomical biases while identifying a medial-caudal ‘hotspot’ of glutamatergic neurons most strongly inhibited by the SNr. In contrast, the GPe strikingly avoids the cholinergic PPN neurons and weakly, but selectively, inhibits a subpopulation of caudal GABAergic and glutamatergic neurons. We proposed that the differential inhibition of regionally-biased, mixed-cell type PPN subpopulations can alter behavioral outputs. By stimulating these inhibitory basal ganglia axons over the PPN, we show that the SNr evokes place aversion while the GPe evokes place preference and decreases locomotion. Together, our findings show that the SNr and GPe mediate opposing valence processing outcomes through the PPN and support a non-canonical basal ganglia motor pathway in which the GPe decreases locomotion through its projections to the PPN.

Star Methods text

Key resources table

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
goat anti-choline acetyltransferase (ChAT)	Millipore	Cat#: AB144P, RRID: AB_2079751
Streptavidin, Cy5	Invitrogen	Cat#: SA1011
Streptavidin, DyLight™ 405	Invitrogen	Cat#: 21831
Donkey anti-goat IgG (H+L) Cross-Adsorbed Secondary Antibody, Alexa Fluor™ 647	Invitrogen	Cat#: A-21447, RRID: AB_2535864
Sheep anti-tyrosine hydroxylase (TH)	Novus Biologicals	Cat# NB300-110, RRID: AB_10002491
DyLight™ 405 AffiniPure™ Donkey Anti-Sheep IgG (H+L)	Jackson ImmunoResearch Laboratories	Cat# 713-475-147, RRID: AB_2340740
Bacterial and virus strains		
AAV1-hSyn-hChR2(H134R)-EYFP	Addgene (Deisseroth Lab)	Cat#: 26973
AAV1-Ef1a-double floxed-hChR2(H134R)-EYFP-WPRE-HGHpA	Addgene (Deisseroth Lab)	Cat#: 20298
AAV1-hSyn-EGFP	Addgene (Roth Lab)	Cat#: 50465
Chemicals, peptides, and recombinant proteins		
D-AP5	Tocris and HelloBio	Cat#: 0106 and HB0225
NBQX	Tocris and HelloBio	Cat#: 1044 and HB0443
CNQX disodium salt	HelloBio	Cat#: HB0205
SR 95531 hydrobromide (GABA _A zine)	Tocris and HelloBio	Cat#: 1262 and HB0901
Experimental models: Organisms/strains		
ChAT-cre mice: B6.129S-Chat ^{tm1(cre)Lowl/MwarJ}	Jackson	RRID: IMSR_JAX:031661
Vgat-cre mice: B6J.129S6(FVB)- Slc32a1 ^{tm2(cre)Lowl/MwarJ}	Jackson	RRID: IMSR_JAX:028862
Vglut2-cre mice: B6J.129S6(FVB)- Slc17a6 ^{tm2(cre)Lowl/MwarJ}	Jackson	RRID: IMSR_JAX:028863
Ai9-tdTomato mice: B6.Cg-Gt(ROSA)26Sor ^{tm9(CAG-tdTomato)Hze/J}	Jackson	RRID: IMSR_JAX:007909
DAT-cre mice [B6.SJL-Slc6a3 ^{tm1.1(cre)Bkmn/J}	Jackson	RRID: IMSR_JAX:006660
C57BL/6J	Jackson	RRID: IMSR_JAX:000664
Software and algorithms		
Igor Pro	WaveMetrics	Version 9.00
GraphPad Prism	Dotmatics	Version 9.5.0
Fiji software	Open source on GitHub	Version 1.51n
ANY-maze® software	Stoelting Company, Wood Dale, IL.	Version 7.36
Clampex software	Molecular Devices	Version 11.2
Other		

Multiclamp™ 700B amplifier		
Axon™ Digidata® 1550B		
Leica SP8AOBS++		
Zeiss Axio Imager Z2		
Sutter Instrument Model P-97		

Resource availability

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Dr. Rebekah C. Evans (re285@georgetown.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- All data and code supporting the current study will be shared by the lead contact upon request.
- No original code was generated in this paper.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Experimental model and study participant details

Animal Welfare

All animal procedures were approved by the Georgetown University Medical Center Institutional Animal Care and Use Committee (IACUC). Measures were taken to ensure minimal animal suffering and discomfort, and protocols were designed to minimize the number of animals used.

Animal Subjects

Homozygous Ai9-TdTomato mice [B6.Cg-Gt(ROSA)26Sor^{tm9(CAG-tdTomato)Hze}/J; JAX# 007909] were bred with homozygous ChAT-Cre mice [B6.129S-Chat^{tm1(cre)Low}/Mwar; JAX# 031661], homozygous Vgat-Cre mice [B6J.129S6(FVB)-Slc32a1^{tm2(cre)Low}/Mwar; JAX# 028862], and Vglut2-Cre mice [B6J.129S6(FVB)-Slc17a6^{tm2(cre)Low}/Mwar; JAX# 028863] to obtain ChAT-Cre/Ai9-TdTomato, Vgat-Cre/Ai9-TdTomato, and Vglut2-Cre/Ai9-TdTomato in-house, respectively. C57BL/6J wildtype mice (Jax# 000664) and DAT-cre mice [B6.SJL-Slc6a3^{tm1.1(cre)Bkmn}/J; JAX#006660] were obtained from Jackson for behavioral experiments. Animals were housed under a 12:12 light-dark cycle (lights on at 06:00 AM) with food and water *ad libitum*.

Experimental Groups

For electrophysiological studies, ChAT-Cre/Ai9-TdTomato, Vgat-Cre/Ai9-TdTomato, and Vglut2-Cre/Ai9-TdTomato mice of both sexes (3 males and 3 females in each group) aged 2-5 months were used. For behavioral studies, C57BL/6J wildtype mice of both sexes, with age-matched littermates randomly assigned to control or experimental groups. DAT-cre mice of both sexes were used for behavioral experiments where specified. Behavioral assessments were exclusively conducted during the light phase, and all testing chambers were cleaned with 70% ethanol between mice to mitigate potential olfactory influences.

Method details

Viral injections and optical fiber implantation surgeries

Mice at least 7 weeks old were briefly anesthetized with inhaled 5% isoflurane using an anesthetic chamber and placed onto a heated pad within the stereotaxic frame (Stoelting 51730UD). The skull was stabilized with evenly positioned ear bars and nose cone properly positioned to deliver continuous 1-3% isoflurane and oxygen at a steady flow of 1L/min throughout the surgery duration. Bupivacaine (5 mg/kg) and carprofen (5mg/kg) were administered as local anesthetic and analgesic, respectively. A small incision was made on the scalp to visualize bregma and lambda, which were used as references to level the skull. Bilateral holes were drilled at either the coordinates of the SNr (AP -3.1 mm, ML $\pm$ 1.4 mm, DV - 4.7 mm relative to bregma) or GPe (AP -0.3 mm, ML $\pm$ 1.9mm, DV -3.9 mm relative to bregma). The 5 μ L Hamilton microsyringe was positioned and 250 nL virus was injected at a rate of 0.2 μ L/min. The syringe was raised and rested 0.5 mm above the injection site for 10 minutes. For *ex vivo* optogenetic experiments, AAV1-hSyn-hChR2(H134R)-EYFP (2.3 x 10¹³ particles per milliliter, Addgene, Cat# 26973) was injected into either the SNr or GPe. For *in vivo* optogenetic experiments, AAV1-hSyn-hChR2(H134R)-EYFP 2.0 x 10¹³ particles per milliliter, Addgene, Cat #26973), AAV1-hSyn-EGFP (1.1 x 10¹³ particles per milliliter, Addgene, Cat# 50465), or AAV1-Ef1a-double floxed-hChR2(H134R)-EYFP-WPRE-HGHpA (2.3 x 10¹³ particles per milliliter, Addgene, Cat#20298) was injected into either the SNr or GPe for experimental and control mice, respectively. For mice used in the *in vivo* optogenetic experiments, an optical fiber (200 μ m core, 0.22 NA, 3.7 mm length) was implanted over the middle-to-rostral PPN (AP -4.5, ML $\pm$ 1.1 mm, DV -3.5 relative to bregma). The skin on either side of the incision site was joined, glued together using VetBond tissue adhesive, and fastened with two wound clips or the optical fiber was fixed in position with dental cement. (C&B-Metabond Quick! Adhesive Luting Cement by Parkell Products Inc, Patterson# 553-3484/CAT# S380; Jet Denture Repair Package by LANG, CAT# 1223F2). Post-surgery, buprenorphine SR (1.5 mg/kg) was administered for long-acting analgesia. Mice were allowed to recover on a heating pad until fully awake and were monitored daily for signs of distress or infection.

Electrophysiological solutions

Acute brain slices were prepared in modified artificial cerebrospinal fluid (aCSF) solutions with final osmolarities ~300-310 mOsm and ~7.4 pH. The slicing solution contained (in mM) 198 glycerol, 2.5 KCl, 1.2 NaH₂PO₄, 20 HEPES, 25 NaHCO₃, 10 glucose, 10 MgCl₂, and 0.5 CaCl₂. The holding solution contained (in mM) 92 NaCl, 2.5 KCl, 1.2 NaH₂PO₄, 30 NaHCO₃, 20 HEPES, 2 MgCl₂, 2 CaCl₂, 35 glucose, 5 sodium ascorbate, 3 sodium pyruvate, and 2 thiourea. Recording aCSF was made up of (in mM): 125 NaCl, 25 NaHCO₃, 3.5 KCl, 1.25 NaH₂PO₄, 10 glucose, 1 MgCl₂, 2 CaCl₂. Whole-cell patch clamp recordings used a potassium methane sulfonate (KMeSO₃)-based internal solution containing (in mM) 122 methanesulfonic acid, 9 NaCl, 9 HEPES, 1.8 MgCl₂, 4 Mg-ATP, 0.3 Tris-GTP, and 14 phosphocreatine for a final osmolarity between 290 and 305 mOsm. The internal solution contained neurobiotin (0.1-0.3%) for post-hoc staining. These reagents enable reliable electrophysiology recordings in adult brainstem neurons^{46,79}.

Slicing and Electrophysiology

Animals ages 2-5 months old were anesthetized with inhaled isoflurane and transcardially perfused with ice-cold slicing solution that had been bubbled with 95% O₂ and 5% CO₂. Mice were decapitated and brains were quickly extracted from the skull, keeping the cerebellum intact. For sagittal slices, the brain hemispheres were separated and the medial side was glued onto a 3% agar block fixed to the stage of a semi-automatic Leica VT1200 microtome. 200 μ m-thick slices were obtained and incubated at 34°C in holding solution for 30 minutes then kept at room temperature. In all steps, the modified aCSF solutions are bubbled with 95% O₂ and 5% CO₂. Slices in recording chamber during whole-cell patch clamp experiments were continuously perfused with oxygenated recording aCSF kept at 28-34°C using a water bath and in-line Warner heater. Cells were visualized using an Olympus OpenStand upright microscope and 565nm Thorlabs LED

light. Recording pipettes with resistance between 1.5–4 M Ω were prepared using borosilicate glass capillaries (World Precision Instruments, Inc. 1B150F-4, 4-inch length, 1.5 mm OD, 0.84 mm ID) with a micropipette puller (Sutter Instrument Model P-97). Recordings were obtained using a MulticlampTM 700B amplifier and AxonTM Digidata[®] 1550B controlled by Clampex 11.2. Voltage-clamp signals were low-pass filtered at 2 kHz and sampled at 10 kHz. Current-clamp signals were low-pass filtered at 10 kHz and sampled at 10 kHz. At the end of recording, the cell was sealed by moving the recording pipette slightly above the soma and an image was taken of the PPN and pipette tip.

***Ex vivo* optogenetic activation**

Whole-field optogenetic activation of channelrhodopsin-infected axons in brain slice was achieved with a blue (470 nm) ThorLabs LED light sent to the tissue via a silver mirror. Light intensity measured at the objective back aperture was 13 mW. Light activation was applied at 20 Hz with 2 ms pulse intervals for 2 seconds. All recordings were conducted in the presence of glutamatergic receptor blockers – 50 μ M D-AP5 (Tocris Cat#0106 and HelloBio Cat#H0225), and 5 μ M NBQX (Tocris Cat#1044 and HelloBio Cat#HB0443) alone or in combination with 20 μ M CNQX (HelloBio Cat#HB0205).

In voltage-clamp, cells were determined to be connected if observable optically evoked inhibitory post synaptic currents (oIPSCs) were measured while holding the cell at –50 mV. In a subset of neurons, 10 μ M GABAzine (Tocris Cat#1262 and HelloBio Cat#HB0901) was applied to ensure the oIPSCs were GABA-a receptor mediated. In a few cells (17/54), small inhibitory currents remained with amplitudes averaging about 24% of the first oIPSC measured. Any cells that required greater than 200 pA to be held at –50 mV or that had access resistance that exceeded 30 M Ω at the start of recording were excluded from oIPSC quantitative analyses as they were likely unhealthy and too leaky; however, the first oIPSC amplitude was used as an approximate measure of input strength in the cell maps (SNr: 21/46; GPe: 11/40). oIPSC amplitudes were measured from the baseline to the peak of each current. The paired pulse ratio (PPR) was calculated from the amplitudes of the first two currents evoked in the 20 Hz train.

For current-clamp recordings, most patched PPN neurons spontaneously fired, but a few required a small amount of hyperpolarizing or depolarizing current to fire [(R)rostral, (C)audal; SNr-PPN_{ChAT} R: 3/14, C: 4/23; SNr-PPN_{Vgat} R: 2/7, C: 9/19; SNr-PPN_{Vglut2} R: 2/13, C: 1/17; GPe-PPN_{ChAT} R: 2/7, C: 9/25; GPe-PPN_{Vgat} R: 2/7, C: 8/18; GPe-PPN_{Vglut2} R: 6/12, C: 7/29 required current). Neurons injected with more than 60 pA were excluded. In this manuscript, we use “pre-stimulation firing frequency” to refer to the time period right before optical stimulation. The pre-stimulation firing frequency was measured using the number of action potentials in the one-second epoch before the start of stimulation. The firing frequency during stimulation was measured using the middle one-second of the 2-second stimulation. The post-stimulation firing frequency was measured in the one-second epoch immediately following the end of the stimulation. The absolute frequency change is the difference between the pre-stimulation and during stimulation frequencies. The rebound frequency change is the difference between the post-stimulation and pre-stimulation frequencies. These values include cells with and without holding current applied. By contrast, the spontaneous firing rate of the cell includes only cells with no holding current applied.

***In vivo* optogenetic activation**

Two behavioral tests, the open field (OF) test and real-time place preference (RTPP), were conducted sequentially to assess gross locomotion and place preference in animals of at least 11 weeks old (mean age $\pm$ SEM; SNr-injected mice: 132 $\pm$ 23.4 days old; GPe-injected mice: 110 $\pm$ 7.4 days old). Optical stimulation was achieved by the application of blue light (473 nm) at 20 Hz with 2 ms pulses (4% duty cycle). Behavioral testing was conducted under both high laser power (4.0–4.5 mW) and low laser power (0.2–0.25 mW) conditions. Low power testing was employed to control for potential behavioral effects related to antidromic stimulation. On day 1, mice underwent OF

testing and were randomly assigned to high or low power stimulation conditions. On day 2 mice were tested in the open field arena for a second time under opposite power conditions. On day 3 mice were tested in the RTPP arena for 10 minutes and were again randomly assigned to high or low power conditions. On day 4, mice were tested in the RTPP arena for 10 minutes without any stimulation to assess retention of place memory. Finally, on day 5, mice underwent RTPP under opposite power conditions (low vs high) and opposite chamber as stimulation zone (stripes vs spots) to those on Day 3.

Open Field Test

OF test was conducted in an opaque arena measuring 40.64 x 40.64 x 40.64 cm. Animals were placed in the center of the arena and allowed to explore freely for 10 minutes. This was followed by 5 minutes of discontinuous bilateral photostimulation at either high or low power, 5 minutes of recovery, another 5 minutes of discontinuous photostimulation at 20 Hz (at either high or low power), and a final 5 minutes of recovery. The discontinuous photostimulation consisted of three 1-minute photostimulation periods interspersed with 1-minute intervals of no photostimulation. Movement during the OF test were recorded and analyzed using ANY-maze® software (Stoelting Company, Wood Dale, IL).

Real-Time Place Preference

The real-time place preference (RTPP) apparatus consisted of a rectangular behavioral arena with three chambers, each measuring 24.60 x 27.94 x 27.94 cm, with a non-reflective grey background. The adjacent chambers had distinct visual cues: one chamber featured stripes and the other spots. These striped and spotted zones were randomly assigned as stimulation zones between animals. The center chamber and one of the adjacent chambers served as neutral zones, while the other adjacent chamber served as a stimulation zone. Animals received unilateral optical stimulation when in the assigned stimulation zone. Laser-on side was randomly chosen and counter-balanced between mice. Mice were also randomly assigned to have low laser power RTPP first or high laser power RTPP first. In each case, mice were given an unstimulated 10-minute trial on the day between the first and second RTPP experiment to ‘unlearn’ which side was stimulated and the second RTPP experiment stimulated the opposite chamber compared to the first RTPP experiment. For example, one mouse would have high power stimulation on the striped side on day 1, no stimulation on day 2, and low power stimulation on the spotted side on day 3. To mitigate locomotor effects with bilateral stimulation, unilateral optical stimulation was used in RTPP. Individual subjects were placed in the RTPP apparatus and allowed 10 minutes to explore both compartments. The time spent in each compartment was recorded in real-time using ANY-maze® software (Stoelting Company, Wood Dale, IL).

Immunohistochemistry and confocal imaging

After electrophysiological experiments, brain slices were fixed overnight in a 4% w/v paraformaldehyde (PFA) solution in phosphate buffer (PB) solution, pH 7.6 at 4°C. The fixed brain slices were then stored in phosphate buffer (PB) solution until immunostaining.

Brain slices were also collected at the conclusion of behavioral experiments. Mice were deeply anesthetized and perfused with phosphate buffer (PB), followed by a fixative solution containing 4% w/v paraformaldehyde (PFA) in PB, pH 7.6, at 4°C. Whole brains were extracted and fixed overnight in the same PFA solution. After fixation, brains were stored in PB solution until further processing. For cubic processing, 200 µm-thick sagittal slices were obtained from post-behavioral whole mouse brains using the PELCO easiSlicer™ Vibratory Tissue Slicer.

A CUBIC tissue clearing protocol⁴³ was combined with immunofluorescence staining as in⁷⁹ for all fixed brain slices from both electrophysiological and behavioral experiments. All steps are performed at room temperature on a shaker. Slices were placed in CUBIC reagent 1 for 1-2 days;

then washed in PB 3x for 1 hour each; placed in blocking solution (0.5% fish gelatin in PB) for 3 hours; placed in primary antibodies for 2-3 days; washed in PB 3x for 2 hours each; placed in secondary antibodies for 2-3 days; washed in PB 3x for 2 hours each; and placed in CUBIC reagent 2 for 2 hours before mounting onto slides (Fisherbrand 12-550-403) in reagent 2 and sealed with frame-seal incubation chambers (Thermo Scientific AB-0577) and a coverslip (Corning 2845-18).

Neurobiotin-filled patched neurons were stained with streptavidin antibody (Cy5, Invitrogen Cat#SA1011, 1:1000 or DyLight™ 405, Invitrogen Cat#21831, 1:1000). Goat anti-ChAT primary antibody (Sigma Cat#AB144P, 1:200) and donkey anti-goat Alexa Fluor™ 647 secondary antibody (Invitrogen Cat#A-21447, 1:333) was used to identify the borders of the PPN. On an example slice to show the SNr injection site, sheep anti-TH primary antibody (Novus Biologicals Cat#NB300-110, 1:1000) and donkey anti-sheep (1:100) was used to delineate the GABAergic SNr from the dopamine-rich TH+ SNc neurons. To identify the approximate location of patched cells for cell mapping, slices were imaged as tiled z-stacks using a Leica SP8AOBS++ at the Microscopy and Imaging Shared Resource core facility at Georgetown University. Additionally, to verify the injection site and optical implant placement in the brain slices from mice that underwent behavioral testing, tiled z-stacks were obtained using either the aforementioned Leica SP8AOBS++ or a Zeiss Axio Imager Z2 at the Georgetown Department of Neuroscience core imaging facility.

Approximate Cell Location

An image of the PPN with the pipette tip still in position was taken at the end of each cell recording and confocal images of the slices after CUBIC clearing were obtained as described above. Confocal images were overlapped with the pipette images using the streptavidin-stained patched cells (which aligned with the pipette tip images) and autofluorescence of the superior cerebellar peduncle (scp) white fibers as landmarks. The perimeter of the PPN was determined by the presence of ChAT+ neurons and generally extended from the edge of the SNr to approximately 2/3 up the scp fibers. ChAT+ neurons were identified by tdTomato expression in the ChAT-Cre mouse line and anti-ChAT staining in the Vgat-Cre and Vglut2-Cre mouse lines. Since there is variability introduced during the slicing process, we matched slices to three sagittal templates representing 1.00, 1.25, and 1.50 mm lateral to the midline in the Paxinos and Franklin's Mouse Brain Atlas 4th edition. The most medial slice (1.00 mm lateral to the midline) characteristically had few to no fibers of the superior cerebellar peduncle at the edge of the slice where the cerebellum would tether. All neurons on this slice were considered 'caudal' following the typical shape of the PPN. Slices matching 1.25 and 1.50 mm from the midline had a region of densely clustered cholinergic neurons and a region of loosely dispersed cholinergic neurons. An oval ROI was created following the length and orientation of ChAT+ neuron spread. The midline of the oval demarcated the rostral and caudal regions. Overlapping the midlines allowed us to evaluate rostral-caudal differences across mice with a small margin of error, particularly along the midline. For this reason, we also mapped the location of patched cells which shows the distance from the midline and can reveal medial-lateral differences. The XY coordinates were determined using Fiji software and plotted in Igor. The color scale applied to the points represented the amplitude of the first oIPSC and ranged from 0 to 250pA so that amplitudes greater than 250pA were shown in the same maximal bright color.

Approximate Implant Location

To verify the location of the optical implant, anatomical landmarks for locating the PPN were identified by immunofluorescent labeled PPN cholinergic neurons. Sections containing visible optical fiber scarring were stained with anti-ChAT immunofluorescent labeling. These histological images were then visually matched to the corresponding sagittal planes of the stereotaxic atlas, based on the identified anatomical landmarks. The approximate position of the optical cannula was localized in the histological images and was in the middle-to-rostral part of the PPN.

Quantification and statistical analysis

Electrophysiological traces were processed in Igor Pro 9 (Wavemetrics) and all statistical analyses were performed in GraphPad Prism 9.5. All parametric data in text is reported as mean $\pm$ standard error of the mean (SEM). Two-tailed unpaired t-tests were used to compare between two groups and one-way ANOVA followed by Dunnett's post hoc test was used to compare three or more groups. If the equality of variances was violated (i.e. the ratio of the largest and smallest standard deviations was greater than 2 and Bartlett test was statistically significant), Welch's ANOVA test was alternatively used. Two-way ANOVA followed by Fisher's LSD post hoc test was performed for percent of pre-optical stimulation frequency as each comparison group was predetermined. Non-parametric data in text is reported as median [interquartile range (IQR)]. Boxplots show medians, 25th and 75th percentile as first and third quartile box edges, and 9th and 91st percentiles as whiskers. We used Mann-Whitney rank-sum test for comparing two groups and Kruskal-Wallis for multiple comparisons. As each comparison group within the Kruskal-Wallis tests was planned and stand alone, uncorrected Dunn's post hoc test was performed to determine significance between groups. To determine if caudal GPe-connectivity is significantly greater than rostral GPe-connectivity, as expected from axon projection patterns, a one-tailed Fisher's exact test and odds ratio effect size was performed. For correlation matrix analyses, non-parametric Spearman r was computed. Statistical details of experiments can be found in the text, figure legends, and Supplemental Table 1. Biological replicates are individual cells (n) from 6 separate mice ($N=3$ males and 3 females) in the electrophysiological experiments and are individual mice of both sexes in the behavioral experiments (numbers described in results text). Same sex littermates were randomly allocated to control or experimental groups and experimenters were not blinded during experiments. Statistical significance was evaluated as $p < 0.05$.

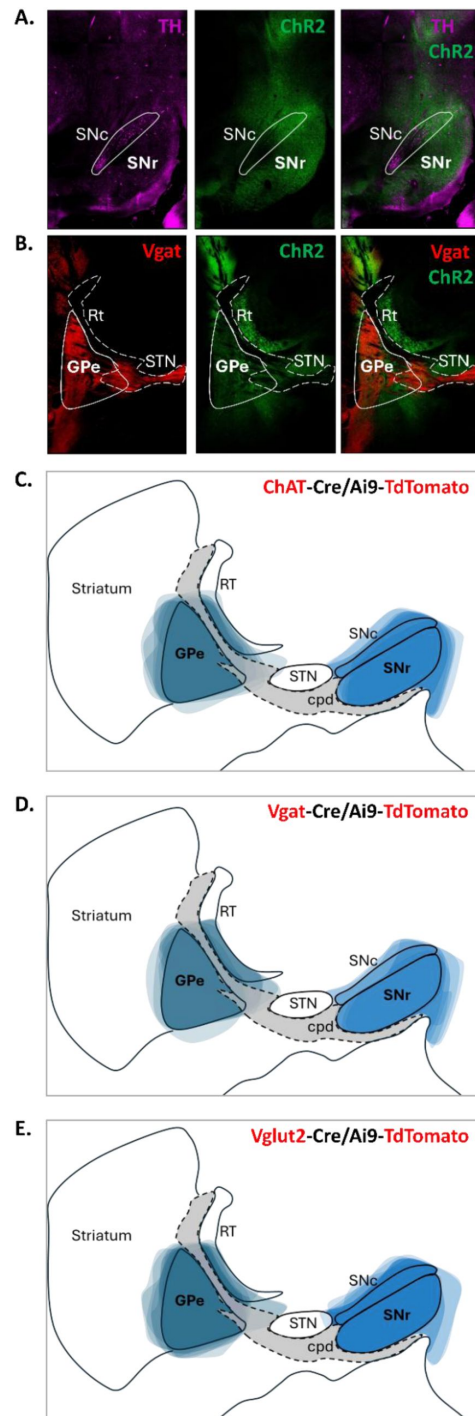


Figure S1.

(A) Representative image of SNr injected with ChR2-EYFP. Cell bodies and axons filled with ChR2 are represented in green. Post hoc staining of tyrosine hydroxylase (TH) to label dopamine neurons in the substantia nigra *pars compacta*. TH+ neurons are represented in magenta. **(B)** Representative image of GPe injected with ChR2-EYFP in a Vgat-Cre/Ai9-TdTomato mouse. Cell bodies and axons filled with ChR2 are represented in green. Vgat+ neurons and their axonal projections are represented in red. **(C-E)** Graphical depiction of virus spread in N=6 SNr and N=6 GPe injected mice in each mouse line: **(C)** ChAT-Cre/Ai9-TdTomato, **(D)** Vgat-Cre/Ai9-TdTomato, and **(E)** Vglut2-Cre/Ai9-TdTomato mice.

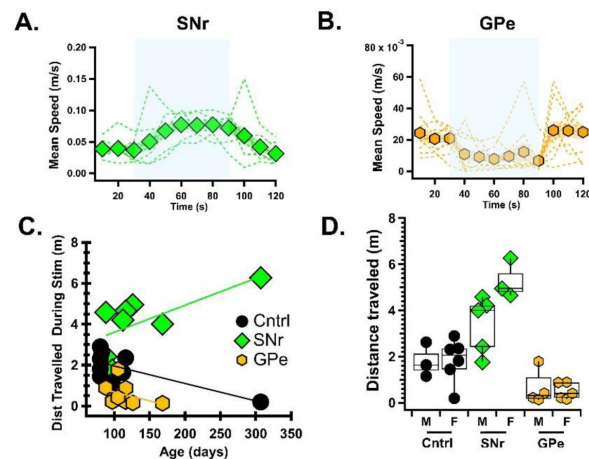


Figure S2.

Speed measurements across age and sex.

(A) Speed of SNr-ChR2 mice during optical stimulation in the open field. Individual mouse traces are shown as green broken lines, with the group average represented by a solid green line with diamond markers. SNr-ChR2 mice exhibited increased speed during stimulation. (B) Speed of GPe-ChR2 mice during six 1-minute optical stimulations in the open field. Individual mouse traces are shown as orange broken lines, with the group average represented by a solid orange line with hexagon markers. GPe-ChR2 mice exhibited a reduction in speed during stimulation. Data are presented as mean $\pm$ SEM. (C) Distance traveled in the open field as a function of age in SNr-ChR2 and control mice. Older SNr-ChR2 mice tend to travel further, whereas older control mice travel less. Individual mouse traces are shown. (D) Comparison of locomotion between male and female SNr-ChR2 mice in the open field, showing a trend toward increased movement in female mice with SNr stimulation.

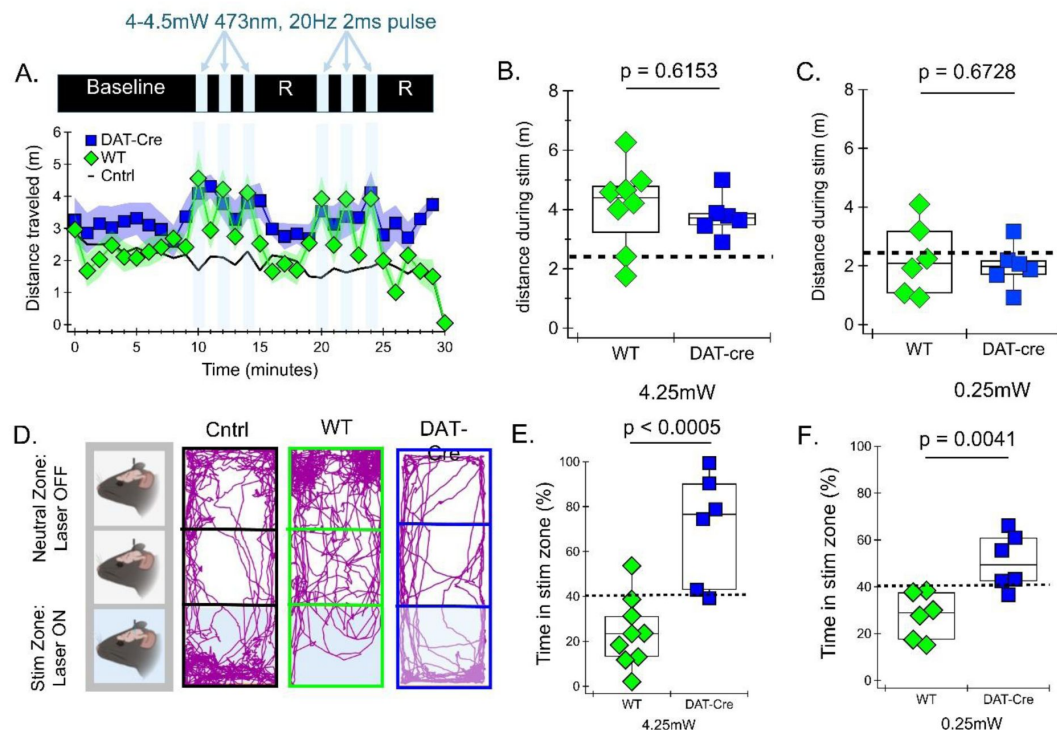


Figure S3.

Opposite place preference despite similar motor responses to nigral axon stimulation in the PPN of DAT-Cre and wildtype mice.

(B) Distance traveled over time in an open field during 1-minute optical stimulations (20 Hz, 4.25 mW) over the pedunculoptine nucleus (PPN). Data are shown for **N** = 6 DAT-Cre mice and wildtype mice injected with Chr2 in the substantia nigra. SNr data from [Figure 7](#). Data are presented as mean $\pm$ SEM.

(C) Average distance traveled for each mouse during stimulation across six 1-minute optical stimulations under high laser power (4.25 mW). (C) Average distance traveled for each mouse during stimulation across six 1-minute optical stimulations under low laser power (0.25 mW). (D) Representative movement traces from individual mice during a real-time place preference task in a three-chamber box. Mice received continuous optical stimulation (20 Hz) of EGFP-or Chr2-expressing axons over the PPN when in the stimulation zone. (E) Percentage of time spent in the stimulation zone under high laser power (4.25 mW) for **N** = 9 wildtype and **N** = 6 DAT-Cre mice injected with Chr2 in the substantia nigra. (F) Percentage of time spent in the stimulation zone under low laser power (0.25 mW) for the same groups.

Figure S4.

Rostral vs caudal implant site alters SNr axon stimulation effects.

(A) Implant location (same as [Figure 7B](#)) with most rostral (purple) and most caudal (red) bilateral implant pairs circled for SNr axon stimulation. (B) Distance traveled in open field in mouse with rostral implants (purple) and mouse with caudal implants (red).

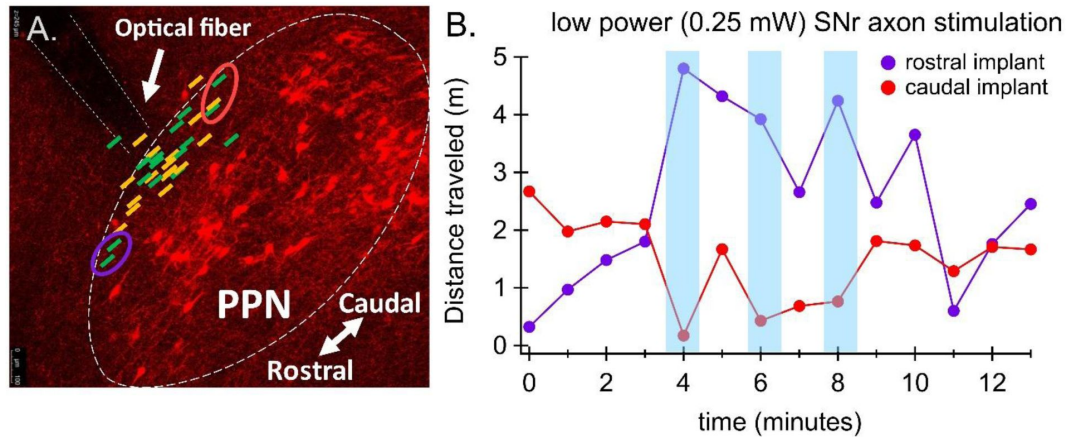


Table 1.

Statistical details of exact p-values and critical values for SNr input to ChAT+ PPN neurons

Parameter	Statistical Test	p-value	ChAT+	
			Rostral	Caudal
oIPSC Amplitude	unpaired t-test, $t=0.9041$, $df=33$	0.3725	$n=15$, 92.6 ± 18 pA	$n=20$, 115.2 ± 17 pA
PPR	unpaired t-test, $t=0.1082$, $df=33$	0.9145	$n=15$, 0.96 ± 0.04	$n=20$, 0.97 ± 0.05
Avg membrane potential	Mann Whitney, $U=87$	0.4706	$n=11$, -54.24 (-56.94 to -50.20) mV	$n=19$, -53.00 (-55.38 to -48.84) mV
% Pre-Opto Frq	Repeated Measures 2way ANOVA Stimulation: $F(1, 35)=81.9$ Region: $F(1, 35)=0.002052$ Interaction: $F(1, 35)=3.273$	<0.0001 0.9641 0.0790	$n=14$ During: 35.62 ± 8.869 % Post: 136.1 ± 7.803 %	$n=23$ During: 52.11 ± 8.432 % Post: 119.1 ± 4.614 %
Δ Frq During Opto	Mann Whitney, $U=131$	0.3602	$n=14$, -2.47 (-3.37 to -1.44) Hz	$n=23$, -2.10 (-3.11 to -0.84) Hz
Δ Rebound Frq	Mann Whitney, $U=79$	0.0142	$n=14$, 1.64 (1.00 to 2.17) Hz	$n=22$, 0.83 (0.42 to 1.14) Hz

Table 2.

Statistical details of exact p-values and critical values for SNr input to Vgat+ PPN neurons

Parameter	Statistical Test	p-value	Vgat+	
			Rostral	Caudal
oIPSC Amplitude	Mann Whitney, U=50	0.6005	n=9, 76.5 (27.0 to 209.9) pA	n=13, 59.7 (37.7 to 269.7) pA
PPR	Mann Whitney, U=48	0.5123	n=9, 1.01 (0.88 to 1.16)	n=13, 1.06 (0.96 to 1.15);
Avg membrane potential	unpaired t-test, t=2.468, df=13	0.0282	n=5, -57.91 ± 1.630 mV	n=10, -52.00 ± 1.472 mV
% Pre-Opto Frq	Repeated Measures 2way ANOVA Stimulation: F(1, 24)=44.63 Region: F(1, 24)=0.09642 Interaction: F(1, 24)=0.8841	<0.0001 0.7588 0.3565	n=7 During: 37.91 ± 15.91 % Post: 100.0 ± 14.57 %	n=19 During: 35.62 ± 8.869 % Post: 113.2 ± 5.187 %
ΔFrq During Opto	Mann Whitney, U=55	0.5336	n=7, -5.03 (-5.55 to -1.34) Hz	n=19, -4.29 (-11.02 to -2.94) Hz
ΔRebound Frq	Mann Whitney, U=63	>0.9999	n=7, 0.80 (-1.31 to 2.33) Hz	n=18, 0.59 (-2.35 to 1.47) Hz

Table 3.

Statistical details of exact p-values and critical values for SNr input to Vglut2+ PPN neurons

Parameter	Statistical Test	p-value	Vglut2+	
			Rostral	Caudal
oIPSC Amplitude	Mann Whitney, U=29	0.0035	n=13, 61.6 (19.2 to 113.9) pA	n=13, 313.5 (143.8 to 729.1) pA
PPR	Mann Whitney, U=77	0.7241	n=13, 1.07 (0.86 to 1.38)	n=13, 0.92 (0.89 to 1.02)
Avg membrane potential	unpaired t-test, t=0.3838, df=25	0.7043	n=11, -53.03 ± 1.815	n=16, -52.09 ± 1.601
% Pre-Opto Frq	Repeated Measures 2way ANOVA Stimulation: F(1,28)=133.6 Region: F(1, 28)=0.5214 Interaction: F(1,28)=0.1765	<0.0001 0.4762 0.6776	n=13 During: 32.62 ± 10.21 % Post: 100.3 ± 6.113 %	n=17 During: 24.71 ± 7.297 % Post: 97.48 ± 2.228 %
ΔFrq During Opto	Mann Whitney, U=55	0.0343	n=13, -3.21 (-12.21 to -1.95) Hz	n=17, -14.08 (-20.04 to -6.07) Hz
ΔRebound Frq	Mann Whitney, U=88	0.3627	n=13, 0.37 (-0.027 to 0.48) Hz	n=17, -0.23 (-0.82 to 1.08) Hz

Parameter	Statistical Test	p-value	ChAT+	Vgat+	Vglut2+
oIPSC Amplitude	Kruskal-Wallis, H(2,37)=3.050	0.2176	n=6, 22.6 (9.75 to 50.20) pA	n=19, 24.77 (6.72 to 54.17) pA	n=15, 65.71 (14.50 to 133.4) pA
PPR	Mann Whitney, U=76	0.0206		n=19, 0.97 (0.76 to 1.2)	n=15, 0.81 (0.55 to 0.93)
Avg membrane potential	Kruskal-Wallis, H(2,69)=2.963	0.2273	n=25, -54.16 (-58.05 to -51.10) mV	n=18, -54.28 (-55.47 to -48.77) mV	n=29, -55.32 (-57.97 to -52.28) mV
% Pre-Opto Frq	Repeated Measures 2way ANOVA Stimulation: F(1, 69)=25.46 Cell type: F(1, 69)=4.444 Interaction: F(1, 69)=2.805	 <0.0001 0.0153 0.0674	n=29 During: 92.39 ± 7.59 % Post: 104.10 ± 5.23 %	n=18 During: 57.70 ± 10.11 % Post: 91.81 ± 6.98 %	n=24 During: 64.15 ± 7.63 % Post: 104.7 ± 5.39 %
	Uncorrected Fisher's LSD, df=138 t=2.971 t=3.223 t=0.6177 t=0.1407 t=1.076 t=1.236	 0.0035 0.0016 0.5378 0.8883 0.2838 0.2184	During: ChAT+ vs. Vglut2+: 28.24 (CI 9.446 to 47.03) ChAT+ vs. Vgat+: 34.69 (CI 13.41 to 55.97) Vglut2+ vs. Vgat+: 6.454 (CI -14.21 to 27.11) Post: ChAT+ vs. Vglut2+: -1.337 (CI -20.13 to 17.45) ChAT+ vs. Vgat+: 11.58 (CI -9.702 to 32.86) Vglut2+ vs. Vgat+: 12.92 (CI -7.741 to 33.58)		
ΔFrq During Opto	Kruskal-Wallis, H(2,69)=17.21	0.0002	n=25, -0.16 (-0.90 to 0.03) Hz	n=18, -1.92 (-3.85 to -0.60) Hz	n=29, -0.89 (-3.28 to -0.43) Hz
	Uncorrected Dunn's test	0.0009 0.0002 0.4020	ChAT+ vs. Vglut2+: Z=3.332 ChAT+ vs. Vgat+: Z=3.755 Vglut2+ vs. Vgat+: Z=0.8380		
ΔRebound Frq	Kruskal-Wallis, H(2,69)=17.21	0.6772	n=8, 0.1041 (-0.52 to 0.41) Hz	n=19, 0.07082 (-0.36 to 0.57) Hz	n=7, -0.1421 (-2.58 to 0.71) Hz

Table 4.

Statistical details of exact p-values and critical values for GPe input to ChAT+, Vgat+, and Vglut2+ PPN neurons

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Additional information

Author Contributions

M.F. and R.C.E. conceptualized the experiments; M.F. performed stereotaxic injections, conducted the electrophysiological experiments, and analyzed the data; A.E.S. and K.C.U. performed stereotaxic injections, conducted the behavioral experiments, and analyzed the data; C.B.S. performed stereotaxic injections for pilot experiments; M.F. wrote the initial draft of the manuscript; M.F. and R.C.E. edited and revised the manuscript.

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Additional files

Video S1. Example video of stimulating SNr axons over the PPN in an open field, related to Figure 7. [🔗](#)

Video S2. Example video of stimulating GPe axons over the PPN in an open field, related to Figure 7. [🔗](#)

Video S3. Example video of stimulating SNr axons over the PPN in the striped zone, related to Figure 7. [🔗](#)

Video S4. Example video of stimulating GPe axons over the PPN in the striped zone, related to Figure 7. [🔗](#)

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Reviewer #1 (Public review):

Summary:

Fallah and colleagues characterize the connectivity between two basal ganglia output nuclei, the SNr and GPe, and a the pedunculo pontine nucleus, a brainstem nucleus that is part of the mesencephalic locomotor region. Through a series of systematic electrophysiological studies, they find that these regions target and inhibit different populations of neurons, with anatomical organization. Overall, SNr projects to PPN and inhibits all major cell types, while the GPe inhibits glutamatergic and GABAergic PPN neurons, and preferentially in the caudal part of the nucleus. Optogenetic manipulation of these inputs in the had opposing effects on behavior - SNr terminals in the PPN drove place aversion, while GPe terminals drove place preference.

Strengths:

This work is thorough and systematic characterization of a set of relatively understudied circuits. They build on the classic notions of basal ganglia connectivity and suggest a number of interesting future directions to dissect motor control and valence processing in brainstem systems.

Limitations:

All the cell type recording studies showing subtle differences in the degree of inhibition and anatomical organization of that inhibition suggest a complex effect of general optogenetic manipulation of SNr or GPe terminals in the PPN. It will be important to determine if SNr or GPe inputs onto a particular cell type in PPN are more or less critical for the how the locomotion and valence effects demonstrated here.

<https://doi.org/10.7554/eLife.102308.2.sa3>

Reviewer #2 (Public review):

Strengths:

Fallah et al carefully dissect projections from SNr and GPe - two key basal ganglia nuclei - to the PPN, an important brainstem nucleus for motor control. They consider inputs from these two areas onto 3 types of downstream PPN neurons: GABAergic, glutamatergic, and cholinergic neurons. They also carefully map connectivity along the rostrocaudal axis of the PPN. They provide important and convincing data on PPN connectivity with two important input structures, which will provide a foundation for many future studies. They also consider the behavioral relevance of these different PPN inputs for controlling movement and reinforcement, showing convincing evidence that SNr and GPe inputs have opposing effects on behavior.

Weaknesses:

The optogenetics and behavioral studies are intriguing, although more work will be required to fit these data together into a specific model of circuit function and to distinguish the locomotor and reinforcement effects. Interestingly, stimulation of SNr axons in the rostral vs caudal PPN likely differs (as predicted by slice experiments), indicating an area for future investigation and dissection of pathways.

<https://doi.org/10.7554/eLife.102308.2.sa2>

Reviewer #3 (Public review):

The study by Fallah et al. provides a thorough characterization of the effects of two basal ganglia output pathways, the SNr and the GPe, on cholinergic, glutamatergic, and GABAergic neurons of the PPN. Using a combination of optogenetics-assisted electrophysiology and behavioral assays in genetically defined mouse lines, the authors show that SNr projections broadly inhibit all PPN subtypes along the rostrocaudal axis, whereas GPe projections are mostly restricted to the caudal PPN and predominantly target glutamatergic neurons, with a lesser effect on GABAergic neurons. Activation of these inputs in vivo revealed opposing behavioral effects: SNr stimulation increased locomotion and caused avoidance in the real-time place preference (RTPP) task, while GPe stimulation reduced locomotion and increased time spent in the stimulation zone.

Strengths:

The evidence for functional connectivity between SNr and GPe inputs and specific PPN cell types is solid and highlights a prominent influence of SNr across the PPN. The identification of a GPe projection that selectively targets caudal glutamatergic PPN neurons is unexpected and highly relevant to understanding basal ganglia-brainstem interactions. The study stands out for its systematic cell-type-specific approach and the combination of electrophysiological and behavioral data. Importantly, the authors addressed key concerns from the initial review by performing new analyses and adding important controls:

Motor activity was re-analyzed at higher temporal resolution, revealing more nuanced effects of stimulation (Fig. S2).

The concern that motor effects might confound RTPP performance was mitigated by analyzing unstimulated test sessions, which showed that place preference or aversion persisted in the absence of stimulation (Fig. 7G).

The potential recruitment of SNc dopaminergic projections was directly tested using DAT-Cre mice, confirming that dopaminergic axon stimulation drives locomotion and reward but does not explain the aversive effect seen with broader SNr activation (Fig. S3).

Weaknesses:

While the revised analyses and added data strengthen the conclusions, the interpretation of the behavioral effects remains somewhat limited by the use of RTPP, which can be influenced by motor changes, even with unilateral stimulation. Nonetheless, the additional controls and thorough discussion now acknowledge and address these caveats appropriately.

Some minor clarifying edits would enhance the manuscript's precision and readability, including improvements to terminology, data presentation, figure referencing, and the organization of behavioral and statistical reporting.

Conclusion:

This is a strong and compelling study that provides a detailed and novel characterization of basal ganglia inputs to the PPN and their behavioral relevance. The authors were responsive to reviewer feedback, and the revised manuscript is significantly improved. The findings advance our understanding of how basal ganglia output pathways engage brainstem circuits to modulate locomotion and valence.

<https://doi.org/10.7554/eLife.102308.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Fallah and colleagues characterize the connectivity between two basal ganglia output nuclei, the SNr and GPe, and the pedunculopontine nucleus, a brainstem nucleus that is part of the mesencephalic locomotor region. Through a series of systematic electrophysiological studies, they find that these regions target and inhibit different populations of neurons, with anatomical organization. Overall, SNr projects to PPN and inhibits all major cell types, while the GPe inhibits glutamatergic and GABAergic PPN neurons, and preferentially in the caudal part of the nucleus. Optogenetic manipulation of these inputs had opposing effects on behavior - SNr terminals in the PPN drove place aversion, while GPe terminals drove place preference.

Strengths:

This work is a thorough and systematic characterization of a set of relatively understudied circuits. They build on the classic notions of basal ganglia connectivity and suggest a number of interesting future directions to dissect motor control and valence processing in brainstem systems. We thank the reviewer for these positive comments.

Weaknesses:

Characterization of the behavioral effects of manipulations of these PPN input circuits could be further parsed, for a better understanding of the functional consequences of the connections demonstrated in the ephys analyses.

We have further analyzed our behavioral data to reveal more nuanced functional effects and included these analyses in Figure S2.

All the cell type recording studies showing subtle differences in the degree of inhibition and anatomical organization of that inhibition suggest a complex effect of general optogenetic manipulation of SNr or GPe terminals in the PPN. It will be important to

determine if SNr or GPe inputs onto a particular cell type in PPN are more or less critical for how the locomotion and valence effects are demonstrated here.

This is a really interesting future direction and we have expanded on these points in the discussion in lines 771-772 and 782-785.

Reviewer #1 (Recommendations for the authors):

(1) Overall these are really valuable studies and help set up a number of future directions.

We thank the reviewer for their positive comments.

(2) I don't have many specific suggestions, but more examples of viral targeting and cell type targeting, including potentially some validation of the genetic identity of the cells targeted, could be useful for considering the details of the ephys experiments.

We agree that understanding which exact SNr and GPe neurons go to which exact PPN populations is an important next step and are planning to conduct future experiments investigating these important questions. Others have found that there is minimal overlap between the three cell types within the PPN discussed in this manuscript (Wang and Morales 2009; Yoo et al. 2017; Steinkellner, Yoo, and Hnasko 2019). One important line of future investigations is to look at the specific inputs onto recently identified subsets of the glutamatergic PPN neurons such as Chx10- and Rbp4-expressing neurons (Goñi-Erro et al. 2023; Ferreira-Pinto et al. 2021). We hope to explore the electrophysiological properties and connectivity of these subtypes in future projects.

(3) More discussion of which PPN cell types might be mediating the optogenetic behavioral effects of bulk SNr or GPe terminal stimulation would be useful for connecting the ephys results with the behavior.

We are also interested in the question of which PPN cell type is most critical for mediating the effects observed in bulk terminal stimulation. While the best experiment would be to stimulate the axons projecting to each specific cell type of the PPN, this is not currently possible due to methodological limitations and lack of studies dissecting which SNr and GPe subpopulations project to each cell type of the PPN. However, in future studies, we plan to leverage the ability of AAV1 to jump a synapse along with Cre/Flp viruses and mouse lines to selectively inhibit cholinergic, GABAergic, or glutamatergic PPN neurons that receive GPe or SNr input to elucidate the contribution of each cell type in mediating behavioral changes in movement and valence processing.

To address these important future directions, we have added additional text in the discussion in lines 771-772 and 782-785.

Reviewer #2 (Public review):

Summary:

Fallah et al carefully dissect projections from SNr and GPe - two key basal ganglia nuclei - to the PPN, an important brainstem nucleus for motor control. They consider inputs from these two areas onto 3 types of downstream PPN neurons: GABAergic, glutamatergic, and cholinergic neurons. They also carefully map connectivity along the rostrocaudal axis of the PPN.

Strengths:

The slice electrophysiology work is technically well done and provides useful information for further studies of PPN. The optogenetics and behavioral studies are thought-provoking, showing that SNr and GPe projections to PPN play distinct roles in behavior.

We appreciate the reviewer's positive evaluation.

Weaknesses:

Although the optogenetics and behavioral studies are intriguing, they are somewhat difficult to fit together into a specific model of circuit function. Perhaps the authors can work to solidify the connection between these two arms of the work.

We have expanded on these topics in the discussion.

Otherwise, there are a few questions whose answers could add context to the interpretation of these results:

(1) Male and female mice are used, but the authors do not discuss any analysis of sex differences. If there are no sex differences, it is still useful to report data disaggregated by sex in addition to pooled data.

We have added a supplementary figure (Figure S2) showing distance traveled during optical stimulation for male and female mice.

(2) There is some lack of clarity in the current manuscript on the ages used - 2-5 months vs "at least 7 weeks." Is 7 weeks the time of virus injection surgery, then recordings 3 weeks later (at least 10 weeks)? Please clarify if these ages apply equally to electrophysiological and behavioral studies. If the age range used for the test is large, it may be useful to analyze and report if there are age-related effects.

Thanks for pointing this out, we have clarified this in the methods. 7 weeks is the youngest age at which mice used for electrophysiology were injected, and all were used for electrophysiology between 2-5 months. For behavior, the youngest mice used were 11 weeks old at time of behavior (8 weeks old at injection). Mice in the GPe-stimulated condition were 110 ± 7.4 SEM days old and mice in the SNr-stimulated condition 132 ± 23.4 SEM days old. We have added these details to the revised manuscript in lines 913 and 963-964.

In addition, we have correlated distance traveled at baseline and during stimulation with age for both SN and GPe stimulated conditions. Baseline distance traveled did not correlate with age, but there was a trend toward more movement during stimulation with older mice in the SN axon stimulation group. We have included these plots in supplemental Figure S2.

(3) Were any exclusion criteria applied, e.g. to account for missed injections?

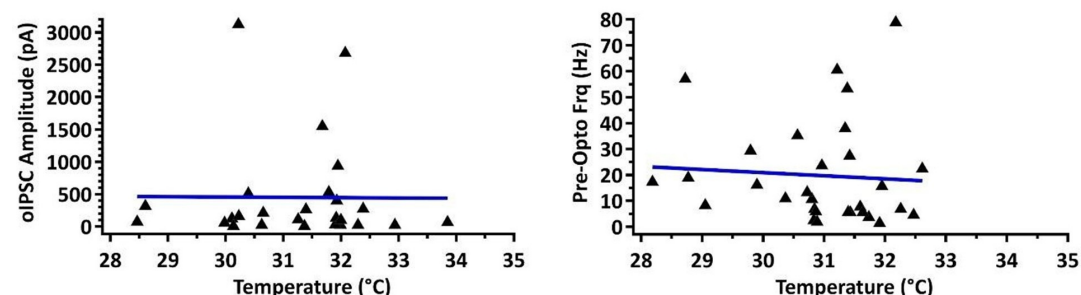
All injection sites and implant sites were within our range of acceptability, so we did not exclude any mice for missed injections or incorrect implant location.

(4) 28-34 degC is a fairly wide range of temperatures for electrophysiological recording, which could affect kinetics.

This is an important consideration, and we agree the wide temperature is not optimal. We have plotted our main measurement of current amplitude in the condition where we found significant differences between rostral and caudal PPN (SNr to Vglut2 PPN neurons) against temperature and found no correlation (Pearson's r value = -0.0076). Similarly, we found no

correlation between baseline (pre-opto) firing frequency and temperature ($r = -0.068$). See Author response image 1.

Author response image 1.

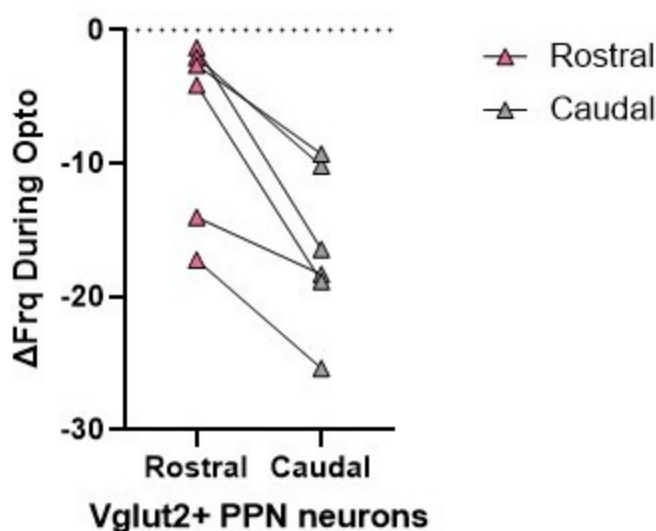


(5) It would be good to report the number of mice used for each condition in addition to $n = \text{cells}$. Statistically, it would be preferable not to assume that each cell from the same mouse is an independent measurement and to use a nested ANOVA.

For electrophysiology, the number of mice used in each experiment was 6 (3 male, 3 female). In the manuscript 'N' represents number of mice and 'n' represents number of cells. Because of the unpredictability of how many healthy cells can be recorded from one mouse, our data were planned to be collected with $n = \text{cells}$, and are underpowered for a nested ANOVA.

However, in many cases, rostral and caudal data were collected from the same mice. While we do not have sufficient paired data for each electrophysiological parameter, analyzing one of our main and most important findings with a paired comparison (with biological replicates being mice) shows a statistically significant difference in the inhibitory effect of SNr axon stimulation on firing rate between rostral and caudal glutamatergic neurons ($p = 0.031$, Wilcoxon signed rank test). See Author response image 2.

Author response image 2.



Reviewer #3 (Public review):

Summary:

The study by Fallah et al provides a thorough characterization of the effects of two basal ganglia output pathways on cholinergic, glutamatergic, and GABAergic neurons of the PPN. The authors first found that SNr projections spread over the entire PPN, whereas GPe projections are mostly concentrated in the caudal portion of the nucleus. Then the authors characterized the postsynaptic effects of optogenetically activating these basal ganglia inputs and identified the PPN's cell subtypes using genetically encoded fluorescent reporters. Activation of inputs from the SNr inhibited virtually all PPN neurons. Activation of inputs from the GPe predominantly inhibited glutamatergic neurons in the caudal PPN, and to a lesser extent GABAergic neurons. Finally, the authors tested the effects of activating these inputs on locomotor activity and place preference. SNr activation was found to increase locomotor activity and elicit avoidance of the optogenetic stimulation zone in a real-time place preference task. In contrast, GPe activation reduced locomotion and increased the time in the RTPP stimulation zone.

Strengths:

The evidence of functional connectivity of SNr and GPe neurons with cholinergic, glutamatergic, and GABAergic PPN neurons is solid and reveals a prominent influence of the SNr over the entire PPN output. In addition, the evidence of a GPe projection that preferentially innervates the caudal glutamatergic PPN is unexpected and highly relevant for basal ganglia function.

Opposing effects of two basal ganglia outputs on locomotion and valence through their connectivity with the PPN.

Overall, these results provide an unprecedented cell-type-specific characterization of the effects of basal ganglia inputs in the PPN and support the well-established notion of a close relationship between the PPN and the basal ganglia.

We thank the reviewer for their positive comments.

Weaknesses:

The behavioral experiments require further analysis as some motor effects could have been averaged out by analyzing long segments.

We have further analyzed our motor effects and included these analyses in supplemental figure S2 in the revised manuscript.

Additional controls are needed to rule out a motor effect in the real-time place preference task.

To address this comment, we analyzed the second day of RTPP, where no stimulation was applied in either chamber. Specifically, we evaluated the time spent in the stimulated chamber during the first minute of the unstimulated RTPP task. We found that the mice that had SNr axon stimulation still avoided the previously stimulated chamber and the mice that had GPe axon stimulation still preferred the previously stimulated chamber. These data have been added to Figure 7 and in the results section lines 564-575.

Importantly, the location of the stimulation is not reported even though this is critical to interpret the behavioral effects.

The implant locations were generally over the middle-to-rostral PPN and we will clarify this in the revised manuscript. These locations are shown in figure 7B.

There are some concerns about the possible recruitment of dopamine neurons in the SNr experiments.

We have added experiments stimulating the SNc dopaminergic neuron axons in the PPN and found very interesting behavioral effects. These are described in more detail below and in the results lines 595-624. These data are also included in Figure S3.

Reviewer #3 (Recommendations for the authors):

(1) Locomotor activity should be analyzed as trial averages instead of session averages. The effect of SNr on locomotion might be showing a rebound of activity in cholinergic neurons, which innervate dopamine neurons and induce locomotion. Furthermore, the variability between animals should be reported, Figure 7C doesn't show a standard deviation.

This is an important point and could reveal different early and late effects of basal ganglia axon stimulation. We have added a time course graph of the trial averages for the distance traveled in the open field with higher temporal resolution (10s vs 1min). This is included in supplemental Figure S2A&B.

The variability between animals was shown as shaded area, but was too light and transparent so it was difficult to see in Figure 7C. We have changed this shading to error bars for better visibility.

(2) SNc projects to the PPN. It has recently been shown that PPN neurons respond robustly to dopaminergic activation, including effects on motor activity (Juarez Tello et al., 2024). The transductions shown in Figure S1 clearly cover to entirety of the SNc. Dopamine blockers should be used in the ex vivo experiments to rule out dopaminergic effects.

This is an important point and one we were particularly interested in as far as the behavioral experiments. We thank the reviewer for bringing this up because it led us to a really interesting result. We have now run an additional experiment using DAT-cre mice and a cre-dependent ChR2 using the same injection site at our constitutive ChR2 experiments. We found that selectively stimulating the SNc dopaminergic axons replicates the increased locomotion at high laser power and replicates the no change in locomotion at low laser power as seen with our constitutive ChR2 experiment. However, the selective dopaminergic axon activation in the PPN is rewarding at both high and low power, while the constitutive ChR2 activation is aversive. We have added these data to supplemental figure S3, and have added text in the results (lines 595-624) and discussion (lines 695-734) about this new exciting finding.

While we can't exclude the possibility of dopamine influence on the electrophysiology experiments (via changes in input resistance or channel properties), the fast synaptic currents measured are uncharacteristic of inhibitory D2 receptor currents (which would be slow), and are inhibited by the GABA_A receptor blocker, GABAzine.

(3) Activation of glutamatergic neurons in the caudal PPN elicits locomotion while the same stimulation in the rostral PPN terminates locomotion. In line with this, the authors report important differences in glutamatergic neurons in the rostral vs caudal PPN (Fig. 5). For the behavioral experiments, the location of the optic fiber is not reported. This is essential for the interpretation of the behavioral experiments. Based on the recent

literature, inhibiting glutamatergic neurons in the rostral and in the caudal PPN will produce opposing effects.

We absolutely agree the rostral and caudal PPN differences are functionally important. In Figure 7B, we have mapped the location of the optical fiber tip for each experiment. Our implant location was generally in the rostral-middle part of the PPN and we have added this to the methods section of the revised manuscript in lines 887 and 1048. While we did not have many implant locations that were specifically rostral or specifically caudal, we did evaluate the behavioral response for our most rostrally-located implant and our most-caudally located implant in the SN axon stimulation experiment. We found that low-power laser activation of nigral axons in the most rostral implant resulted in increased locomotion but in the most caudal implant resulted in decreased locomotion. This increased locomotion exactly what we would expect when rostral PPN neurons (that normally inhibit movement) are preferentially inhibited, and decreased locomotion is what we would expect when caudal PPN neurons (that normally promote movement) are inhibited. Future experiments using more precise rostral and caudal implant locations will be needed to fully parse out the functional role of rostral vs caudal PPN. See Figure S4 (two green implant sites are circled for one mouse because the implants were bilateral).

(4) Even though the authors made an effort to dissect out the motor component during the RTPP task, this was not entirely achieved. Low laser power was still able to decrease activity following GPe stimulation, causing the animal to spend more time in the stimulated compartment. It is not clear the reason for using RTPP as opposed to CPP, which will not have the confound of the effects on motor activity. The interpretation of these data is problematic.

This is an important consideration, and the reviewer is correct that we can't completely eliminate a motor contribution to our RTPP experiment. We attempted to minimize potential motor confounds by utilizing unilateral stimulation and our supplemental videos show that the mice can escape the stimulated chamber.

However, to address this comment, we analyzed the second day of RTPP, where no stimulation was applied in either chamber. Specifically, we evaluated the time spent in the stimulated chamber during the first minute of the unstimulated RTPP task. We found that the mice that had SN stimulation still avoided the previously stimulated chamber and the mice that had GPe axon stimulation still preferred the previously stimulated chamber. These data have been added as Figure 7G and in the results section lines 564-575.

(5) The resting membrane potential for cholinergic, glutamatergic, and GABAergic neurons is not reported.

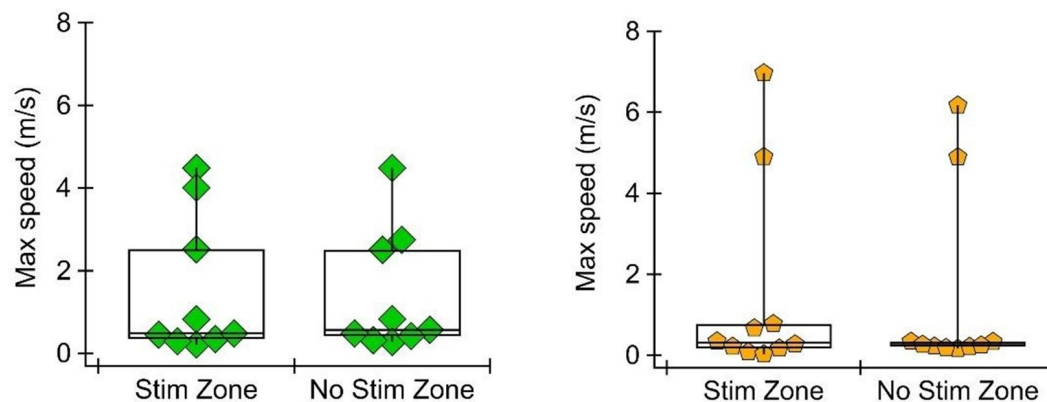
Since a majority of PPN neurons are spontaneously active, we have reported the average membrane voltage during the pre-optical stimulation period in supplementary table 1.

(6) During the RTPP, the animals were stimulated unilaterally with the purpose of reducing the optogenetic effects on locomotion, but no data support this claim. Please report the locomotor measurements during unilateral stimulation.

To address this comment, we have analyzed the speed of the mouse in each compartment (stimulated vs non-stimulated) during the RTPP task. We found that the mean speed does differ, in the direction expected (i.e., mice are on average slower in the GPe stimulated zone where they spend more time, and mice are on average faster in the SNr stimulated zone where they spend less time). This is expected because when the mouse spends more time in a zone, it is more likely to spend time grooming or staying still, but it could still be evidence of

motor response to the stimulation. To evaluate how fast the mouse is able to move with and without unilateral stimulation, we measured maximum speed in the stimulated and unstimulated zone. We found that maximum speed does not differ between stimulated and unstimulated zones in either the SNr or GPe group. See Author response image 3.

Author response image 3.



(7) Given the similarity of the parameters evaluated for all three PPN cell types, the results could be presented in a table, it will be easier to summarize.

This is a good point and we have added supplemental tables 1-4 for key electrophysiological findings.

(8) The text is repetitive in some parts.

We have gone through the results to edit out repetitive text. For example, lines 244-260 and 274-287 have been rewritten for clarity and efficiency.

(9) Lines 609-620: the behavioral effects after SNr stimulation are not mediated by the PPN, please correct.

We have corrected this.

(10) The number of patched GABAergic neurons in the caudal PPN is almost double the number of patched neurons in the rostral PPN. This contrasts with the high density of GABAergic neurons in the rostral PPN reported in the literature, and therefore, the probability of recording GABAergic neurons will be much higher in the rostral PPN. Please comment.

It is true that there are more GABA neurons in the rostral region, but on a sagittal slice, the rostral region occupies a smaller area compared to caudal and there is a notable cluster of GABAergic neurons in the caudal region (Mena-Segovia et al. 2009). The number of visible and healthy cells with obvious fluorescence against background fluorescence in the heavily myelinated tissue of the PPN is unpredictable and it is possible that the dense number of GABA neurons in the rostral region conglomerates the fluorescence of individual cell somas, making it difficult to detect as many rostral neurons. While we did our best to equally patch rostral and caudal neurons based on our best judgment during the experiment, neurons were ultimately designated as 'rostral' or 'caudal' after post-hoc staining for the cholinergic neurons, as described below.

(11) Describe how the rostral and caudal PPN regions were defined and how the authors ensured consistency across recordings.

We have added more details about the definition of rostral vs caudal PPN in to the methods in lines 1042-1053.

(12) Please report the proportion of GABAergic neurons showing STD vs STP for rostral and caudal PPN. The data in Figure 3 might be averaging out some important differences. Figure 3L suggests some differences in the proportions.

The variability within the GABAergic population was really interesting and we plan to pursue this in the future. We have defined STD as $PPR < 0.95$ and STP as $PPR > 1.05$ and added the proportions of caudal and rostral GABAergic PPN neurons with each type of short-term synaptic plasticity to lines 253-257.

(13) Please report whether the mice's compartment preferences during the habituation were taken into account for the selection of the laser-on compartment.

Mice were not habituated to the chamber in the unstimulated condition prior to the RTPP experiment. Laser-on side was randomly chosen and counter-balanced between mice. Mice were also randomly assigned to have low laser power RTPP first or high laser power RTPP first. In each case, mice were given an unstimulated 10-minute trial on the day between the first and second RTPP experiment to 'unlearn' which side was stimulated and the second RTPP experiment stimulated the opposite chamber compared to the first RTPP experiment. For example, one mouse would have high power stimulation on the striped side on day 1, no stimulation on day 2, and low power stimulation on the spotted side on day 3. This is now explained more thoroughly in lines 564-575 and lines 992-998.

(14) Some references to figure panels are missing in the text.

We have carefully reviewed the manuscript to ensure figure panels are referenced in the text.

(15) The interpretation in lines 724-725 is not supported by the data given that GPe inputs to cholinergic neurons are negligible.

We have reworded much of the discussion.

(16) Some parts of the discussion should go into the "ideas and speculation" subsection of the discussion.

We have rewritten sections of the discussion.

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